Quantum Mechanics

Lectures 1–11 — Prof. V. Balakrishnan

Handwritten lecture notes, transcribed

Contents

1 Lecture 1: Quantum Mechanics (Introduction)

(in Q.M)

$*$ Math is simpler than classical physics.

$*$ Every thing at ultimate level turns out to obey Q.M.

$*$ Q.M applies to everything i.e. photons, electrons, humans, galaxy. It is just that its manifestations become dramatic when we observe small object, & become more dramatic if small objects are moves with very high speed.

$*$ The fact that two solid object does not penetrate one another is due to Q.M. principle (pauli exclusion principle).

$*$ magnetism, superconductor, electric conduction, propogation of sounds in solids, these are all quantum phenomenas.

'The phenomena of diamagnetism, paramagnetism depends on quantum mechanical principales, as they are not explainable from classical physics.'

$*$ Black body radiation is also explained from Q.M.

$*$ uncertainty principle.

The position and canonical conjugate momentum of any particle can not be measured simultaneously and accuratly (to arbitrary precision).

$$\Delta x \cdot \Delta p \ \gtrsim \ \hbar$$

uncertainty principle is intrinsic in nature :-

It is not lack of resolution / experimental lack, it is that the position and momentum are not defined to be measured to with arbitary accuracy (when measured simultaneously).

$$\Delta r \cdot \Delta p_r \ \geq \ \hbar/2$$

$$\Delta y \cdot \Delta p_y \ \geq \ \frac{\hbar}{2}$$

$*$ we can see that hamiltonian formalism is translated to Quantum mechanics.

The whole assumption is that we have

– Hamitonian and

– Hamiltonian framework then we translate to Q.M.

Q.) It is very deep and harder question to ask how to quantize or how do we do Q.M. for a system which are not classical hamiltonian in classical limit.

e.g. friction in system.

(Q) *"Problem of dissipation in Q.M is still an open problem."*

classically, we know that

$$\{x, p_x\} = 1$$

Q.M

$$\Delta x \cdot \Delta p \ \geq \ \frac{\hbar}{2} \qquad \text{where } \hbar = h/2\pi$$

$$\Delta x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \sqrt{\text{variance}}$$

in Q.M $\delta x = \sqrt{\langle x^2\rangle - \langle x\rangle^2}$ is calculated w.r.t quantum mechanical probability distribution. and that distribution will be specified through the Shröndiger's equation.

For any two arbitrary canonically conjugate dynamical variable $\exists$ uncertainty principle.

$$(\Delta A)\cdot(\Delta B) \;\geq\; \tfrac{1}{2}\big|\langle[A,B]\rangle\big|$$

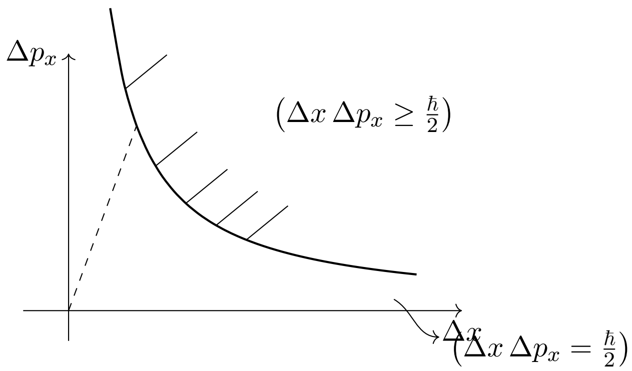
$$[A,B] = AB - BA$$

$\langle[A,B]\rangle$ = expectation value of $[A,B]$.

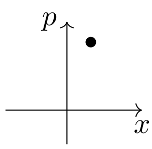
we know that

$$\{x, p_y\} = 0 \qquad \text{— classical}$$

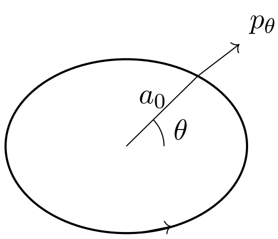
$$(\Delta x\cdot\Delta p_y \geq 0) \qquad \text{— Quantum Physics}$$



Now, the idea of phase space point and trajectory is no more in Q.M.



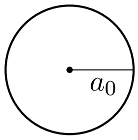
The idea of orbit :-



Here, θ, p_θ can be measured with arbitary precision

so, we can not use orbits to describe motion of $\bar{e}$ in an atom.

Also, Orbital theory is not completely correct.



n	ℓ	m
$\underline{1}$	0	0

If any e^- is orbiting with radius a_0, its angular momentum can not be zero. but it is told that

$$\frac{\sqrt{\ell(\ell+1)}\ h}{2\pi} = 0$$

All objects in Q.M are treated with state vector which is generalization of wave function.

ex.

– e^- is a state vector

– photon is a state vector.

– Even the molecular orbital theory is not correct (s, p, d, f) [n orbitals also] are not correct.

"we should not extrapolate this categorisation of objects into waves & particles to the microscopic domain". It is conceivable that we have objects which have properties of both waves & particles, & it is also conceivable that $\exists$ some objects which has properties of either of them, depending upon how we measure them."

2 Lecture 2

The state of a quantum system is not specified by a point in phase-space but rather it is specified by an element of linear vector space. i.e. vector "state vector". $(|\psi\rangle) \Big/ |\Psi(t)\rangle$.

$\rightarrow$ This state vector is an element of LVS, and all the required information can be extracted from just state vector. and this state vector evolves in time.

$\rightarrow$ we have to throw the idea of representing system as variation of many dynamical variables (independent), $x_1, x_2, x_3 - x_n$ along with idea of phase space.

Digression for LVS

Linear vector space :-

Contains set of elements ("vectors"). $|\psi\rangle, |\phi\rangle, |\chi\rangle \in V$ with following properties / definition.

- $|1\rangle + |2\rangle = |3\rangle \in V$; closure
- $|1\rangle + (|2\rangle + |3\rangle) = (|1\rangle + |2\rangle) + |3\rangle$; Addition is associative
- $|1\rangle + |2\rangle = |2\rangle + |1\rangle$; Add$^{\text{n}}$ is commutative
- $a|\psi\rangle \in V \hookrightarrow$ fields can be scalar Real/complex $(a, b, c, \cdots \in F(C \text{ or } R))$
- $a\,(b|1\rangle) = (ab)|1\rangle \in V$
- $a\,(|1\rangle + |2\rangle) = a|1\rangle + a|2\rangle$
- $\exists$ Null vector such that $|1\rangle + |0\rangle = |1\rangle$

- $\exists$ an additive inverse such that $|1\rangle + |-1\rangle = |0\rangle$

[The page of the notebook carrying the definition of the dual space and the inner product is not present in the scans; only its show-through is visible. The text resumes below.]

$$\langle\phi|\psi\rangle^* = \langle\psi|\phi\rangle$$
$$\left(\langle\phi|\psi\rangle^T\right)^\dagger = \langle\psi|\phi\rangle^* \quad \Rightarrow \quad \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$$

The Norm of a vector :-

$$\|\psi\| \stackrel{\text{def}}{=} \langle\psi|\psi\rangle^{1/2} > 0\,; \quad = 0 \ \text{ iff } \ |\psi\rangle = |0\rangle$$
$$\|\psi + \chi\| \ \leq \ \|\psi\| + \|\chi\| \qquad \text{(triangle inequality)}$$
$$\left|\langle\phi|\psi\rangle\right|^2 \ \leq \ \langle\phi|\phi\rangle\,\langle\psi|\psi\rangle \qquad \text{(cauchy–Schwarz inequality)}$$
$$\begin{cases} \bar{a}\cdot\bar{b} = |a||b|\cos\theta \\ \cos\theta \leq 1 \Rightarrow \bar{a}\cdot\bar{b} \leq |a||b| \end{cases}$$

$\hookrightarrow$ equals only when

$$|\phi\rangle = a|\psi\rangle$$

3 Lecture 3

Linear independence :-

if $|\phi_1\rangle, |\phi_2\rangle, \cdots |\phi_n\rangle$ are linearly independent

then if

$$\sum_i a_i|\phi_i\rangle = 0$$

so sol$^{\text{n}}$ becomes $a_i = 0 = a_1 = a_2$

Span :-

A set of vectors $|\phi_1\rangle, |\phi_2\rangle \cdots |\phi_n\rangle$ are said to span the LVS, iff any vector can be written as linear combination of these vectors. $(|\phi_1\rangle -, |\phi_n\rangle)$.

example

(1) $\hat{e}_x$ & $\hat{e}_y \longrightarrow$ L.I , do not span $\mathbb{R}^3$

(2) $\hat{e}_x, \hat{e}_y, \hat{e}_z \rightarrow$ L.I, span $\mathbb{R}^3$ (orthogonal set) (unit vectors)

(3) $\hat{e}_x, \hat{e}_y, \hat{e}_z, \hat{e}_x + \hat{y} \longrightarrow$ span, not L.I

(4) $\hat{e}_x,\ \hat{e}_y + \hat{e}_x,\ \hat{e}_z + \hat{e}_y + \hat{e}_x \longrightarrow$ L.I, span $\mathbb{R}^3$ (oblique set of coordinate axis vectors) (not-unit vector)

Basis :-

A set of vectors which are linearly independent & span the LVS forms a basis set in the LVS.

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}$$

For orthogonal basis $\delta_{ij} = 0$ if $i \neq j$

Diamentionality :-

The no. of basis vectors defines the diamentionality of LVS.

$\therefore$ dim $V = \#$ of vectors in basis set.

$\longrightarrow$ Infinite dimentional LVS does not have finite basis

$\mathbb{R}^2$ is 2-d ; $\mathbb{R}^3$ is 3-D LVS, $\cdots$ $\mathbb{R}^n$ is n-dimensional LVS.

Here is what can go wrong for infinite-diamensional LVS.

for $\mathbb{R}^n$:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} = |\phi\rangle$$

$$\langle \phi | \phi \rangle = \sum_{i=1}^{N} |x_i|^2 = \|\phi\|^2$$

[we could have defined $\|\phi\|$ as $\left(\sum_{i=1}^{N} |x_i|^p\right)^{1/p}$, but for $p = 2$ only l_p is self dual. dual of $l_p = l_q$ such that $\frac{1}{p} + \frac{1}{q} = 1$ (for $p = q = 2$, $l_p = l_q$)]

for $N \to \infty$ there is no gaurentee that $\langle \phi | \phi \rangle$ converges / have finite value.

so we have to make this piece (length of vector) to be finite.

$$\sum_{i=1}^{N} |x_i|^2 < \infty$$

[$|x_i|^2$ should vanish before $n \to \infty$ (i.e. $n^{-1} \to 0$) $\Rightarrow$ $|x_i|$ should vanish faster than $\frac{1}{\sqrt{n}}$]

$$|f\rangle = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = \sum_{i=1}^{N} f_i |i\rangle \;\Rightarrow\; \langle f | f \rangle < \infty \;\Rightarrow\; \sum_{i=1}^{N} |f_i|^2 < \infty$$

linear vector space of square summable sequences. , l_2

ex.,

(1)

$$(x_1, x_2, x_3 \cdots x_\infty) = \left(1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{4}} \cdots\right)$$

$\hookrightarrow$ diverges. not an element of l_2.

(2)

$$x_r = \frac{1}{r^\epsilon} \qquad \text{for, } \epsilon > \tfrac{1}{2} \text{ it is in } l_2$$

$[|\phi\rangle = (x_1, x_2 \cdots x_r)$ is in l_2 if $x_r = \frac{1}{r^\epsilon}$

$\langle\phi|\phi\rangle = (x_1^2 + x_2^2 + \cdots x_r^2)$, $\sum_{r=1}^{\infty}(x_r)^2 \to$ converges, $\sum_r \frac{1}{r^{2\epsilon}} \to$ converges, $2\epsilon > 1 \Rightarrow (\epsilon > \frac{1}{2})]$

(3)

$$x_n = \frac{\ln n}{n^{0.6}} \quad \text{in } l_2$$

$$|x_n|^2 = \frac{(\ln n)^2}{n^{1.2}} \in l_2$$

also $\frac{(\ln n)^{100}}{n^{0.6}} \in l_2$

(log is weaker than power & power is weaker than exponential)

Normalization :-

$$\frac{|\psi\rangle}{\sqrt{\langle\psi|\psi\rangle}} = \frac{|\psi\rangle}{\|\psi\|}$$

Gram – Schmidt orthonormalization :-

finding two orthonormal basis vectors.

let us say we have any $|\psi_1\rangle, |\psi_2\rangle$ two basis vectors

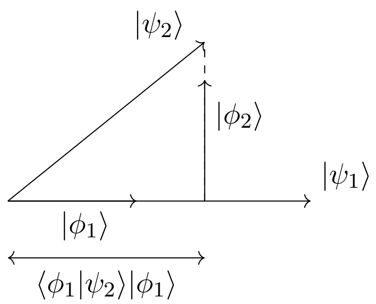
we can produce $|\phi_1\rangle$ & $|\phi_2\rangle$ such that $\langle\phi_i|\phi_j\rangle = \delta_{ij}$.

$|\phi_1\rangle$ = one of orthonormal vector would be simply $\dfrac{|\psi_1\rangle}{\|\psi_1\|}$

$$|\phi_2\rangle = \frac{|\psi_2\rangle - \langle\phi_1|\psi_2\rangle|\phi_1\rangle}{\|\xi\|} = |\phi_2\rangle$$

This will gaurantee that

$$\langle\phi_1|\phi_2\rangle = \boxed{0}$$



$\hookrightarrow$ or

$$\Big\{|\phi_1\rangle\langle\phi_1|\Big\}|\psi\rangle = \langle\phi_1|\psi\rangle\,|\phi_1\rangle$$

$$(n \times 1)(1 \times n) \;\longrightarrow\; (n \times n)$$

(projection operator)

let us say

$$|\psi\rangle = v_1|\phi_1\rangle + v_2|\phi_2\rangle + v_3|\phi_3\rangle$$

magnitude projection along $|\phi_1\rangle \hookrightarrow (\langle\phi_1|\psi\rangle = v_1)$.

projection of $|\psi\rangle$ along $|\phi_1\rangle =$

$$\Big(|\phi_1\rangle\langle\phi_1|\Big)\psi\rangle = |\phi_1\rangle v_1 = v_1|\phi_1\rangle$$

$\hookrightarrow$ projection operator. (of $(n \times n)$ matrix identification)

[In LVS in addition to linear vectors we have linear operators & these operators acts upon these vectors to produce another vector.]

$$\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k} \qquad \boxed{P = \hat{i}\hat{i}\ (\hat{j}\hat{j})\ \hat{k}\hat{k}}$$

$$(\hat{i}\hat{i})\vec{v} = \vec{v}\cdot\hat{i}\,\hat{i} = v_1\hat{i} \quad \text{or} \quad \hat{i}v_1$$

$\hookrightarrow$ dyadic in tensor algebra / or tensor product

(projection operator)

$$P_n = |\phi_n\rangle\langle\phi_n|$$

$$P_n^2 = |\phi_n\rangle\langle\phi_n|\phi_n\rangle\langle\phi_n|$$

(1 unity iff $\|\phi_n\| = 1$)

$$= |\phi_n\rangle\langle\phi_n|$$

$$\left(\sum_n |\phi_n\rangle\langle\phi_n|\right)|\psi\rangle = 1\cdot|\psi\rangle$$

so,

$$\sum_n |\phi_n\rangle\langle\phi_n| = \text{unit operator.}$$

we know that,

$$P_n^2 = P_n \qquad \text{(Idempote matrix) i.e. for matrix } M;\ M^\alpha = M \text{ for any } \alpha \in Z$$

$$P_n^2 - P_n = 0$$

$$P_n(P_n - \not{\Bbb{1}}) = 0 \qquad \text{eigen value} = (0, 1)$$

$\hookrightarrow$ caushy relation $\lambda(\lambda - 1) = 0$ will be satisfied by matrix itself.

These are two very important properties of LVS.

1) orthonormality

$$\langle\phi_i|\phi_j\rangle = \delta_{ij}$$

2) completeness.

$$\sum_{n=1}^{n} |\phi_n\rangle\langle\phi_n| = \not{\Bbb{1}}$$

ex.

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

[$|\phi_1\rangle$ & $|\phi_2\rangle$ also provides basis for operators]

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}$$

$$|\phi_1\rangle\langle\phi_1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|\phi_2\rangle\langle\phi_2| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sum_{i=1}^{2} |\phi_i\rangle\langle\phi_i| = |\phi_1\rangle\langle\phi_1| + \langle\phi_2|\phi_2\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$|\phi_1\rangle\langle\phi_2| = \text{operator} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$|\phi_2\rangle\langle\phi_1| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

we know that,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{cases} a = a_{11} \\ b = a_{12} \\ c = a_{21} \\ d = a_{22} \end{cases}$$

$$= a_{11}|\phi_1\rangle\langle\phi_1| + a_{12}|\phi_1\rangle\langle\phi_2| + a_{21}|\phi_2\rangle\langle\phi_1| + a_{22}|\phi_2\rangle\langle\phi_2|$$

$$\begin{aligned} a_{11} &= \langle\phi_1|A|\phi_1\rangle \\ a_{12} &= \langle\phi_1|A|\phi_2\rangle \\ a_{21} &= \langle\phi_2|A|\phi_1\rangle \\ a_{22} &= \langle\phi_2|A|\phi_2\rangle \end{aligned} \qquad A_{nm} = \langle\phi_n|A|\phi_m\rangle$$

Any vector $|\psi\rangle$ can be uniqly expanded in form

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle$$

Any operator A can be expanded in form ; where $\{|\phi_n\rangle\langle\phi_m|\}$ forms **basis**.

$$A = \sum_{n,m} A_{nm}\, |\phi_n\rangle\langle\phi_m|$$

4 Lecture 4

l_2 : space of square summable sequence

$$\sum_{i=1}^{\infty} |x_i|^2 < \infty$$

$$\Big(l_2(a,b)\Big) = \int_a^b |f(x)|^2 \, dx \quad (< \infty)$$

necessary cond$^{\text{n}}$ for $f(x)$ to be l_2 is that

$$\Big\{ \lim_{x \to \pm\infty} f(x) \longrightarrow 0 \Big\}$$

why not $(x \to b, a)$?

$\hookrightarrow$ it should die to zero sufficiently rapidly

$\lim_{|x|\to\infty} f(x) \longrightarrow 0$ should go faster than $\frac{1}{\sqrt{|x|}}$ (?) **How.**

$\to$ we will require wave function to be bounded.

Let's look at space of square integrable functions. $\to l_2(-1,1)$

where,

$L_1 \to$ space of integrable functions.

$l_2 \to$ space of square integrable fun. (self dual)

$$l_2(-1,1) = \int_{-1}^{1} dx \, |f(x)|^2 < \infty$$

[let us find basis set for any function. so that we can expand it in the given basis]

let $|\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle \cdots \to x^0, x^1, x^2, \cdots$ forms basis. (orthonormal)

$f_0(x) = c_0$

whats

$$\int_{-1}^{1} dx \ c_0^2 = 1 \qquad \Rightarrow \qquad c_0 = \frac{1}{\sqrt{2}}$$

$$f_1(x) = ax + b \qquad \text{(in general).}$$

find a, b such that

$$\begin{cases} \langle \phi_0 | \phi_0 \rangle = 1 \\ \langle \phi_1 | \phi_1 \rangle = 1 \\ \langle \phi_0 | \phi_1 \rangle = 0 \end{cases}$$

$$\Rightarrow \int_{-1}^{1} dx \ f_1(x) f_0(x) = 0$$

$$\Rightarrow \int_{-1}^{1} (ax+b) \, \frac{1}{\sqrt{2}} \, dx = 0$$

$$\Rightarrow \left[\frac{ax^2}{2} + bx \right]_{-1}^{1} = 0$$

$$\left(\frac{a}{2}+b\right)-\left(\frac{a}{2}-b\right)=0$$
$$2b=0 \quad \Rightarrow \quad \boxed{b=0}$$

So
$$f_1(x)=ax$$

so normalizing
$$\int_{-1}^{1} f_1(x)^2\,dx=1$$
$$\int a^2x^2\,dx=1$$
$$\frac{a^2}{3}\left(x^3\right)\Big|_{-1}^{1}=1$$
$$\frac{a^2}{3}(1+1)=1$$
$$\boxed{a=\left(\frac{3}{2}\right)^{1/2}}$$
$$f_2(x)=ax^2+bx+c$$
$$\int_{-1}^{1} dx\ f_2(x)f_0(x)=0$$
$$\int dx\ f_2(x)f_1(x)=0$$
$$\int dx\ \left(f_2(x)\right)^2=1$$

three cond$^{\mathrm{n}}$ are sufficent to find a,b,c.

Let us Redefine Normalization condition to get rid of weird constants.

lets call these functions as $P_n(x)$ (Legendre polynomials)

we have,
$$\int_{-1}^{1} dx\ P_n(x)P_m(x)=\frac{2}{2n+1}\delta_{nm}=\frac{2}{2m+1}\delta_{mn}=\frac{2\ \delta_{mn}}{\sqrt{(2n+1)(2m+1)}}$$

if
$$P_0(x)=1 \qquad P_1(x)=x, \qquad P_2(x)=\frac{3x^2-1}{2}$$

Redefining normalization condition for legendre polynomials.

Even the fourier transform (series) is really writing a function in some basis set. Any periodic fun. can be written as (expanded in fourier series). where '$\cos nx$' and '$\sin nx$' are basis set for state vector $f(x)$.
$$f(x+\beta)=f(x)$$
$$f(x)=\sum_{n=0}^{\infty}\left(a_n\cos nx+b_n\sin nx\right) \qquad (n=0,1,2,3\cdots)$$

a_n, b_n are coefficients of expansion

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

[**Generalized normalized condition**

$$\int_{-\infty}^{\infty} d\mu(x)\ \phi_n^*(x)\phi_m(x) = A_n\delta_{nm} \qquad (A_n \to \text{weight factor})$$

$d\mu(x) \to$ measure

ex.

$$\int_{-\infty}^{\infty} dx\, e^{-x^2} \phi_n^*(x)\phi_m(x) = A_n\delta_{nm}$$

$\hookrightarrow$ $\phi_n(x) =$ Hermite polynomials

$d\mu(x) \to$ measure is used to make sure that $\int_{-\infty}^{\infty} d\mu(x)\phi_n^*(x)\phi_n(x) < \infty$ remains finite.

for $d\mu(x) = dx\, e^{-x}$, $\phi(x)$ forms Laguerre polynomial.]

If $f(x)$ is not periodic, we can expand $f(x)$ in fourier integral

$$f(x) = \int_{-\infty}^{\infty} dk\ e^{ikx}\ \tilde{f}(k)$$

for orthonormality

$$\langle\phi_n|\phi_m\rangle = 0\ , \quad \Longleftrightarrow \quad \int_a^b f_n^*(x)f_m(x)\, dx = \delta_{mn}$$

for completeness

$$\sum_{n=1}^{\infty} |\phi_n\rangle\langle\phi_n| = 1$$

$$\sum_n f_n^*(x)f_n(x') = \delta(x-x')$$

$$\int dx'\ \delta(x-x')\,\psi(x') = \psi(x)$$

$\to$ unit operator

[so in this way we can define our own polynomials. let, $d\mu(x) = dx \cdot x^2$, $\phi_n(x)$ forms Vishal's polynomial. $\smile$]

If $\{|\phi_n\rangle\}$ is an orthonormal basis in any LVS, any ket $|\psi\rangle \in V$ can be uniquely expanded as

$$|\psi\rangle = \sum_{n=1}^{N} c_n|\phi_n\rangle \qquad (n\to\infty) \text{ for LVS of infinite diamensions.}$$

where,

$$c_m = \langle\phi_m|\psi\rangle$$

If we have different basis say $|\chi_n\rangle$

$$|\psi\rangle = \sum_{n=1}^{N} d_n|\chi_n\rangle = \sum_i d_i|\chi_i\rangle \qquad \cdots ①$$

where

$$d_j = \langle \chi_j | \psi \rangle \qquad \cdots ②$$

To find relation between c_m and d_i

we can expand $|\chi_i\rangle$ in $|\phi_n\rangle$ basis

$$|\chi_i\rangle = \sum_m h_{mi} |\phi_m\rangle$$

$$\Rightarrow \quad h_{mi} = \langle \phi_m | \chi_i \rangle$$

So, eq$^{\text{n}}$ ① becomes

$$|\psi\rangle = \sum_i^N d_i |\chi_i\rangle = \sum_i \sum_m d_i \, h_{mi} \, |\phi_m\rangle$$

or

$$\sum_n \left(\sum_i h_{ni} d_i \right) |\phi_n\rangle$$

So

$$\boxed{c_n = \sum_i h_{ni} d_i}$$

$$c_n = \langle \phi_n | \psi \rangle = \sum_i \langle \phi_n | \chi_i \rangle \langle \chi_i | \psi \rangle$$

$$= \langle \phi_n | \left(\sum_i |\chi_i\rangle\langle\chi_i| \right) |\psi\rangle$$

($\rightarrow$ unit operator)

$$= \langle \phi_n | \psi \rangle$$

So the whole point is any time we take new basis we automatically insert the identity (unit operator).

$$\langle \psi | = \sum_{n=1}^N c_n^* \langle \phi_n | \ , \qquad |\psi\rangle = \sum_{n=1}^N c_n |\phi_n\rangle$$

$$\langle \psi | \psi \rangle = \sum_{n=1}^N c_n^* c_n \langle \phi_n | \phi_n \rangle = \sum_n |c_n|^2 < \infty$$

as, $\langle \phi_n | \phi_n \rangle = \underline{1}$

we can expand any function $f(x)$ on basis set formed by polynomials (Legendre, Laguerre polynomials, hermite polynomials and fourier series (if $f(x)$ is periodic). If $f(x)$ is not periodic then we can expand it in continuous basis $\{e^{ikx}\}$ labelled by 'k'.

$$f(x) = \int_{-\infty}^{\infty} dk \; e^{ikx} \, \tilde{f}(k)$$

$\tilde{f}(k)$ = fourier transform of $f(x)$.

[$e^{ikx} \sim$ unit vectors / basis set. $\tilde{f}(k)$ are like components in e^{ikx} direction / basis.]

for $l_2(-1,1)$

$$f(\theta) = \sum_{\ell=0}^{\infty} c_\ell \, P_\ell(\cos\theta)$$

$$c_\ell = \tfrac{1}{2}\int_{-1}^{1} d(\cos\theta) \; P_\ell^*(\cos\theta) f(\theta)$$

$$\{P_\ell(\cos\theta) \text{ is real so, } P_\ell^*(\cos\theta) = P_\ell(\cos\theta)$$

$\rightarrow$ comes from orthogonality relation

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} dk \; e^{ik(x-x')} = \delta(x-x')$$

Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} dk \; \tilde{f}(k)\, e^{ikx} \;, \qquad \tilde{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dx \; e^{-ikx} f(x)$$

5 Lecture 5

l_2, $L_2^{(a,b)}$ both are infinite diamensional LVS.

Idea of cauchy sequence :-

let us assume we have set of complex no.

$\{z_n\}$, then for convergence of sequence.

$$|z_n - z_m| \longrightarrow 0 \qquad \text{as, } n, m \rightarrow \infty$$

[$\{z_1, z_2, z_3 \cdots z_n \cdots\}$ formed according to some rule.]

This implies

also ($\Rightarrow \lim_{n\to\infty} z_n$ exists) $\lim_{n\to\infty} z_n =$ some complex no.

If we include the limit point of sequence $z_n (n \rightarrow \infty)$, then $\{z_1, z_2 \cdots z_n\}$ is called as closed set. (or closed sequence).

$\Rightarrow \{|\psi_n\rangle\}$ is a cauchy sequence if $\left(\lim_{n,m\to\infty} \left\|\psi_n - \psi_m\right\| \longrightarrow 0\right)$

$\Rightarrow$

$$\lim_{n\to\infty} |\psi_n\rangle = |\psi\rangle$$

limiting vector

if $|\psi\rangle \in V$ then it is called **complete vector space**

i.e. If the limit vector of every cauchy sequence in a LVS (V) is also in V, then V is a complete LVS.

A complete LVS with an inner product is a Hilbert space.

$*$ The reason we need such sequence is to make sure that operations with infinite diamensional vectors do not lead to errors.

$*$ A Hilbert space with a denumerable basis $\{|\phi_n\rangle\}$ $(n = 1, 2, 3 \cdots \text{ inf})$ [$\rightarrow$ (able to count)] is called a *separable Hilbert space.*

[$\hookrightarrow$ we will be working in separable hilbert space in whole **Q.M**]

$$\Rightarrow \quad |\psi\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle$$

$$c_n = \langle \phi_n | \psi \rangle \qquad \text{if } \{|\phi_n\rangle\} \text{ forms orthonormal basis.}$$

(assuming basis are orthonormal)

$$\langle \phi_n | \phi_m \rangle = \delta_{mn}$$

$$\langle \psi | \psi \rangle = \sum_{n=1}^{\infty} |c_n|^2 = \|\psi\|^2 < \infty \qquad \text{(finite)} \in l_2.$$

$$\begin{cases} \displaystyle\sum_{n=1}^{\infty} |\phi_n\rangle\langle\phi_n| = \mathbb{1} \\ \langle \phi_n | \phi_m \rangle = \delta_{mn} \end{cases}$$

Linear operators

operators acts on $|\psi\rangle$, where linear operator acts linearly

$$\begin{cases} \hat{A}\big(|\psi\rangle + |\chi\rangle\big) = A|\psi\rangle + A|\chi\rangle \\ \hat{A}\big(c|\psi\rangle\big) = c\,\hat{A}|\psi\rangle \\ (\hat{A} + \hat{B})|\psi\rangle = \hat{A}|\psi\rangle + \hat{B}|\psi\rangle \\ \hat{A}\big(\hat{B}|\psi\rangle\big) = (AB)|\psi\rangle \end{cases} \qquad (A, B \text{ are linear operators})$$

in general $AB \neq BA$ operators do not commute.

Postulates

(1) $\rightarrow$ Every dynamical system is described by a "state vector" $|\psi\rangle$ which is element of a Hilbert space.

where system can be

– 1 particle

– 2 particle

– 10 particle

– Human being $\cdots$ galaxy

$\rightarrow$ every one of such system is described by their "own" state vector.

$\rightarrow$ Every information is burried in this vector $\underline{|\psi\rangle}$.

(2) $\rightarrow$ $|\psi\rangle$ is fun. of time generally $\underline{|\psi(t)\rangle}$.

As system evolves $|\psi(t)\rangle$ changes with time and rule of evolution describes the change

(3) $\rightarrow$ Physical measurables are associated with "**self adjoint**" operators. i.e. $(\hat{L}, \hat{p}, x, \vec{v}, \cdots$ **every observable**).

For every physical observable we have corresponding operator. not necessary independent with every; for ex. FOR k.E only velocity / momentum operator are will work.

To associate values with operator, we refer to eigen values of operators.

for operator

$$A|\psi\rangle = \lambda|\psi\rangle$$

then 'λ' is eigen value of A with eigen vector $|\psi\rangle$.

For a given eigen value, if we have two or more linearly independent eigen vectors, then eigen values are repeated eigen values / degenerate eigen value.

It is also possible that no. of eigen values are infinite if LVS is infinite diamensional.

For $n \times n$ matrix it can have max of $n-$ eigen values.

(4) Eigen values are in some sence "the only measurable values" for an operator cooresponding to physical observable should be real.

$\rightarrow$ If $A = A^\dagger$ (Hermitian conjugate) (All eigen values are real for operator A.)

i.e. if operator is self adjoint it's guaranteed that its all eigen-values are real.

Note :– self adjoint are not equally same as Hermitian conjugate except in case of matrices.

Note : we can have contineous set of eigen values in case of infinite diamensional LVS.

For $\underline{x}$, position of a particle can take any value for a definite momentum.

(4) The result of the measurement for any physical measurable observable is always *an* real eigen value for the corresponding operator.

{keep in mind every eigen value will be real for physical observable operators.}

Note :– We can not say which eigen value will appear on measurment because of quantum uncertainty laws.

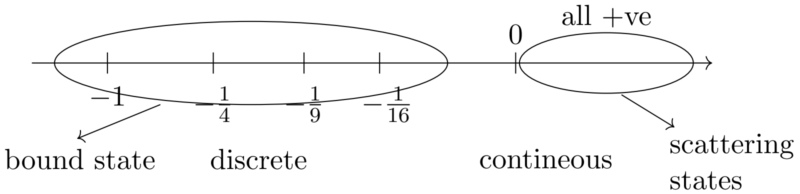
On the other hand we can repeatedly make measurments, take the average we call that mean value of the specific observable. or expectation value i.e. $\underline{\langle x\rangle}$. and we have rule in Q.M to calculate expectation value. It also gives rule to calculate scatter about mean.

This can be achieved by making infinite identical copies of system and make measurements on each of them and take mean. of $\langle x\rangle$.

Average may or may not be an eigen state of that operator.

$*$ spectrum of an operator is the set of all eigen values of the operator. These eigen values can be contineous / discrete eigen values.

ex. the energy of e^- in H like atom goes like



For Any operator A, which acts upon $|\psi\rangle$ state of system

let $\lambda_1, \lambda_2 \cdots$ be the eigen values of A with eigenvectors $|\phi_1\rangle, |\phi_2\rangle, \cdots$

assuming $|\phi_i\rangle$ are orthonormal basis $\langle\phi_i|\phi_j\rangle = \delta_{ij}$

$$|\psi(t)\rangle = \sum_{n=1}^{\infty} c_n|\phi_n\rangle = \sum_{n=1}^{\infty} c_n(t)\,|\phi_n\rangle$$

$\hookrightarrow$ assuming operators are time independent (explicitly).

$$c_n(t) = \langle\phi_n|\psi\rangle$$

$\hookrightarrow$ probability amplitude saying the actual state is $|\phi_n\rangle$.

so c_n tells us how much of $|\psi\rangle$ points along $|\phi_n\rangle$.

$\{|c_n(t)|^2$ is the probability that the system is in the state $|\phi_n\rangle$ at time t.

if $\langle\psi|\psi\rangle = 1$

$$\Rightarrow \quad \sum_n |c_n|^2 = \underline{1}$$

6 Lecture 6

$$\langle A\rangle = \frac{\sum_n \lambda_n |c_n|^2}{\langle\psi|\psi\rangle} \Bigg/ \sum_n |c_n|^2$$

$$A|\phi_n\rangle = \lambda_n|\phi_n\rangle$$

$$\langle\psi(t)|\psi(t)\rangle = \sum_n |c_n|^2 = \underline{1}$$

$$= \frac{\sum_n \lambda_n \langle\psi|\phi_n\rangle\langle\phi_n|\psi\rangle}{\langle\psi|\psi\rangle}$$

$$= \frac{\langle\psi\,|\sum_{n=1}^{\infty}\lambda_n|\phi_n\rangle\langle\phi_n|\,\psi\rangle}{\langle\psi|\psi\rangle}$$

$$= \left\langle \psi \,\middle|\, \sum_{n=1}^{\infty} A|\phi_n\rangle\langle\phi_n \,\middle|\, \psi \right\rangle \Bigg/ \|\psi\|^2$$

$$= \frac{\langle\psi\,|A\sum_{n=1}^{\infty}|\phi_n\rangle\langle\phi_n|\,\psi\rangle}{\langle\psi|\psi\rangle}$$

$$\sum_{n=1}^{\infty} |\phi_n\rangle\langle\phi_n| = 1$$

$$\boxed{\langle A\rangle = \frac{\langle\psi|A|\psi\rangle}{\langle\psi|\psi\rangle}} \qquad \text{or} \qquad \boxed{\langle A\rangle = \frac{\langle\Psi(t)|A|\Psi(t)\rangle}{\langle\Psi|\Psi\rangle}}$$

[$\langle A\rangle$ is diagonal matrix element of $\hat{A}$ in basis formed from $\{|\Psi(t)\rangle\}$.]

for a normalized $|\psi\rangle$, $\langle\psi|\psi\rangle = 1$

So,

$$\boxed{\langle A\rangle(t) = \langle\psi|A|\psi\rangle}$$

where $|\psi\rangle$ is time dependent $|\Psi(t)\rangle$.

classical Physics / Quantum Physics

classical Physics

- Dyn. variables in **phase space** : $q_1, q_2 \cdots \; ; p_1, p_2 \cdots$
- PB relations

$$\{q_i, p_j\} = \delta_{ij}$$
$$0 = \{q_i, q_j\} = \{p_i, p_j\}$$

$H(q, p)$

$$\left.\begin{aligned} \dot{q}_i &= \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= \frac{-\partial H}{\partial q_i} \end{aligned}\right\} \text{dynamics of variables}$$

such that,

$$\{q_i(t), p_j(t)\} = \delta_{ij}$$
$$\{q_i(t), q_j(t)\} = 0$$
$$\{p_i(t), p_j(t)\} = 0$$

Quantum Physics

- Dynamical variables are replaced by self adjoint operators (acting upon the vectors in the **hilbert space**).
- Phase space $\xrightarrow{\text{replaced}}$ Hilbert space.
- PB's are replaced by commutator operators.

$$[A, B] = AB - BA$$
$$\frac{[q_i, p_j]}{i\hbar} = \frac{q_i p_j - p_j q_i}{i\hbar} = \delta_{ij}$$
$$[q_i, q_j] = 0 = [p_i, p_j]$$

Lec 6 (contd.) – Why we need it in the relation ?

Lec 6 > [Why we need $i\hbar$ in $\frac{[q_i,p_j]}{i\hbar} = \delta_{ij}$ relation ?]

◊ Commutator becomes anti-hermitian.

$$A = A^\dagger$$

$$B = B^\dagger$$

$$[A, B]^\dagger = (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = -[A, B] = [B, A]$$

So

$$[A, B]^\dagger = -[A, B]$$

since $AB - BA$ is also a physical measurable, so we have to make sure its eigen values are real, which suggests we have to make it hermitian.

$$\frac{[A,B]}{i\hbar} \text{ has real eigen values.}$$

'$\hbar$' is for diamentional reasons. such that $\frac{[q_i,p_j]}{i\hbar}$ becomes diamensionless.

Now we know that

$$\langle A\rangle(t) = \frac{\langle\psi(t)|A|\psi(t)\rangle}{\langle\psi(t)|\psi(t)\rangle}$$

The Shrodinger Equation :-

The,

$$\boxed{i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle}$$

{operator eq$^{\text{n}}$ for vectors in Hilbert space

where H is an operator.

(postulate :-) *statement :-*

The time rate of change of state vector $|\psi(t)\rangle$ is given by action of Hamiltonian operator on the state vector.

Note :-

$|\psi(t)\rangle$ can be uniquely determined if we have initial state vector $|\psi(0)\rangle$.

Since 'H' is time independent, it appears that Shrodinger equation is autonomous in nature.

$|\psi(t)\rangle$ can be realized by a coloumn vector where as H can be realized by a squarre matrix in Hilbert space.

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$$

$$\frac{d}{dt}|\psi(t)\rangle = -\frac{iH}{\hbar}|\psi(t)\rangle$$

$$\boxed{|\psi(t)\rangle = \psi(0)\, e^{-\frac{iHt}{\hbar}}|\psi(0)\rangle}$$

$$\boxed{|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}}|\psi(0)\rangle}$$

$e^{-\frac{iHt}{\hbar}}$ is operator acts on $|\psi(0)\rangle$

Suppose (t_0) is starting / initial time.

solution becomes

$$\int_{t_0}^{t}\frac{d\big(|\psi(t)\rangle\big)}{|\psi(t)\rangle} = \int_{t_0}^{t}\frac{-iH}{\hbar}\,dt$$

$$\boxed{|\psi(t)\rangle = e^{-\frac{iH(t-t_0)}{\hbar}}\,|\psi(t_0)\rangle}$$

$$\langle\psi(t)| = \langle\psi(t_0)|\, e^{\frac{iH^\dagger(t-t_0)}{\hbar}}$$

($\rightarrow$ operator)

as eigen values of H to be real, H is Hermitian i.e. $\underline{H = H^\dagger}$

$$\langle\psi(t)| = \langle\psi(t_0)|\, e^{\frac{iH(t-t_0)}{\hbar}}$$

so,

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(t_0)|\, e^{\frac{iH(t-t_0)}{\hbar}} \cdot e^{-\frac{iH(t-t_0)}{\hbar}}\, |\psi(t_0)\rangle$$
$$= ?$$
$$\begin{cases} e^A \cdot e^B \neq e^{A+B} & \text{for } [A,B] \neq 0 \\ e^A \cdot e^{-A} = e^0 = \mathbb{1} \text{ operator} \end{cases}$$

$\hookrightarrow$ true always. as $\{A, -A\} = 0$

$$= \langle\psi(t_0)|\mathbb{1}|\psi(t_0)\rangle$$
$$\boxed{\langle\psi(t)|\psi(t)\rangle = \langle\psi(t_0)|\psi(t_0)\rangle}$$

conservation of norm of state vector.

statement :

The analogue of conservation of phase space volume element is reflected in conservation of norm of state vector $|\psi(t)\rangle$. It is because of fact that the Hermitian conjugate of operator $e^{\frac{iH(t-t_0)}{\hbar}}$ is its inverse.

lets call

$$e^{-\frac{iH(t-t_0)}{\hbar}} = U(t, t_0) = \text{evolution operator}$$

or the time development operator. (unitary). or propogator.

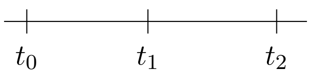
Such operator $U(t, t_0)$ has interesting properties.

$$U^\dagger = U^{-1}(t, t_0)$$

(or)

$$UU^\dagger = U^\dagger U = I \qquad \text{— unitary matrix}$$

If system has t_1 time as initial time between t_0 and t_2



$$U(t_2, t_0) = U(t_1, t_0)\, U(t_2, t_1)$$

semi – group property.

Q) If find $f(t)$.

$$\frac{d}{dt} f(t) = H(t) f(t)$$
$$\Rightarrow \quad \frac{df(t)}{f(t)} = H(t)\, dt$$

$$\Rightarrow \quad \boxed{f(t) = e^{\int_0^t H(t')\,dt'}\; f(0)}$$

$\rightarrow$ Not the formal sol^{n} as

$$e^{H(t_1)+H(t_2)} \neq e^{H(t_1)} \cdot e^{H(t_2)}$$

$$\big(H(t_1), H(t_2)\big) \neq 0$$

7 Lecture 7 :-

operators in Q.M are repesented as $\hat{A}$.

$*$ A particle moving in one diamension with no potential.

1) Analog of H in Q.M

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

Shrodinger eq^{n} becomes.

$$i\hbar \frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\psi(t)\rangle = \frac{\hat{p}^2}{2m}|\Psi(t)\rangle$$

$$|\Psi(t)\rangle = e^{-\frac{i\hat{H}t}{\hbar}}\,|\Psi(0)\rangle$$

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{1}$$

Just as we take coordinate system / axis for (x, p), In Q.M. we must choose some basis. (in terms of which every other vector is expressed).

$*$ Most basic physical measurable quantity is $\hat{x}, \hat{p}$ everything else is derived.

For convenience we should choose the basis set formed from eigen vector of operator (which is associated with each some physical measurable of system).

lets use some basis called as position basis.

$$\{|x\rangle\} \text{ such that}$$

$$\hat{x}|x_0\rangle = x_0|x_0\rangle$$

$\hat{x} =$ operator

$|x_0\rangle =$ eigen state ($|x_0\rangle$ is ket corresponding to a particle with precise location x_0).

$x_0 =$ eigen value

x_0 can take any no. from $-\infty$ to $+\infty$

$$x_0 \in (-\infty, \infty).$$

So it is a contineous basis.

lets choose basis orthonormal to each other.

$$\langle x|x'\rangle = \delta(x - x') = \begin{cases} 1 & x = x' \\ 0 & x \neq x' \end{cases}$$

The completeness relation looks like

$$\int dx \; |x\rangle\langle x| = \mathbb{1} \qquad \text{completeness.}$$

Any state vector can be written as

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty} c_n |x_n\rangle \qquad c_n = \langle x_n|\Psi(t)\rangle$$

but basis are contineous, so we should replace it with integral. (as $|x\rangle$ is contineous basis).

$$|\Psi(t)\rangle = \int dx \; \langle x_0|\Psi(t)\rangle \, |x_0\rangle$$

[we could hv written as, $|\Psi(t)\rangle = \mathbb{1}|\Psi(t)\rangle = \int dx \, |x\rangle\langle x|\Psi(t)\rangle$
$|\Psi(t)\rangle = \int dx \; \langle x|\Psi(t)\rangle \, |x\rangle$]
coefficients are now labled by contineous variable 'x'.

$$\langle x_n|\Psi(t)\rangle = c_x(t) = \Psi(x,t)$$

= probability amplitude that position is 'x' at time t.
$|\psi(x,t)|^2$ = probability density that position lies b/w x, $x + dx$ at time t.
lets drop subscript 'n' as x is contineous.

$$\langle x|\Psi(t)\rangle = \Psi(x,t) = \text{wave function}$$

since it is in position basis, $\Psi(x,t)$ is called as position wave function.
lets choose momentum basis / momentum space set $\{|p_0\rangle\}$

$$\hat{p}|p_0\rangle = p_0|p_0\rangle$$

($\hat{p} \to$ operator, $p_0 \to$ eigen value, $|p_0\rangle$ = eigen vector.)

$$\langle p|p'\rangle = \delta(p - p') \qquad \text{orthogonal } \{|p_0\rangle\} \text{ basis set}$$

$$\sum |p\rangle\langle p| \quad \Rightarrow \quad \int dp \; |p\rangle\langle p| = \mathbb{1} \qquad \text{completeness.}$$

so state vector can be written as.

$$|\Psi(t)\rangle = \int dp \; \langle p|\psi(t)\rangle \, |p\rangle$$

$$\langle p|\Psi(t)\rangle = \tilde{\Psi}(p,t)$$

($\to$ to distinguish from $\Psi(x,t)$)
$\hookrightarrow$ wave function in momentum basis. or momentum space wave function
we can write

$$\tilde{\Psi}(p,t) = \langle p|\Psi(t)\rangle$$

$$= \langle p| \left(\int dx \; |x\rangle\langle x| \right) |\Psi(t)\rangle \qquad (\rightarrow \text{unit operator})$$

$$\tilde{\Psi}(p,t) = \int dx \; \langle p|x\rangle \; \langle x|\Psi(t)\rangle$$

$$\tilde{\Psi}(p,t) = \int dx \; \langle p|x\rangle \; \Psi(x,t)$$

where $\langle p|x\rangle$ is the probability amplitude that when the position of particle is 'x' its momentum is 'p'.

where, $\langle p|x\rangle$ will turn out to be $e^{\frac{ipx}{\hbar}}$

$$\psi(x,t) = \int dp \; \langle x|p\rangle \; \tilde{\Psi}(p,t)$$

Q) which Basis is more convenient out of $|x\rangle$ & $|p\rangle$?

Ans It is easier to work in position basis. because of following reason.

we know that,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

if we use position basis

$$\hat{H}|x_0\rangle = \frac{\hat{p}^2|x_0\rangle}{2m} + V(\hat{x})|x_0\rangle$$

$$= \frac{\hat{p}^2|x_0\rangle}{2m} + V(x_0)|x_0\rangle$$

($V(\hat{x}) \rightarrow$ operator ; $V(x_0) \rightarrow$ eigen value)

once we know how $\hat{p}$ acts on $\{|x_0\rangle\}$ we can easily compute $\hat{H}|x_0\rangle$, but in case of momentum basis $\{|p\rangle\}$.

$$\hat{H}|p_0\rangle = \frac{\hat{p}^2}{2m}|p_0\rangle + V(\hat{x})|p_0\rangle$$

$$= \frac{p_0^2}{2m} + V(\hat{x})|p_0\rangle$$

$V(\hat{x})$ can be complicated function of x, thus will complicate the solution.

for $\hat{H} = \frac{p^2}{2m} + \frac{1}{2}k\hat{x}^2$ simple - Harmonic oscillator.

we can use $\{|x\rangle\}$ and $\{|p\rangle\}$ with same priority.

Q) what is $\hat{p}|x_0\rangle$?

Simultaneous Eigen states :-

Suppose we have two operators A, B such that

$$\begin{aligned} A^\dagger &= A \\ B^\dagger &= B \end{aligned} \qquad \text{i.e. Hermitian operators.}$$

Q.) can I find simultaneous eigen states of A and B. can I find eigen states / eigen functions which are common to both of these operators.

Ans) In general there is no reason we can find such eigen states. But we can expect some eigen states can be common.

If $[A, A^\dagger] = 0$ i.e. if A commutes with its hermitian

such a matrix can always be diagonalised by similarity transformation.

for $A^\dagger = A$ (In all physical operator $A^\dagger = A$)

$$\Rightarrow \quad [A, A^\dagger] = 0$$

we can simultaneously diagonalise A, B if

$$[A, B] = 0 \; ; \quad AB - BA = 0$$

$$\begin{cases} \text{Such that } SAS^\dagger = \text{diagonal } M \\ \qquad\qquad SBS^\dagger = \text{Diagonal } N \end{cases}$$

Here "simultaneously" means we can diagonalise with the same similarity transformation (S).

$\rightarrow$ If $[A, B] \neq 0$

we can not find simultaneous eigen states.

i.e. we can't find complete common set of eigen states. But they may share one or more common eigen-states.

Note :– This does not mean that every eigen-state of A is also the eigen-state of B. if $[A, B] = 0$.

ex. take fun. on a line.

$$\mathbb{1} \, f(x) = f(x)$$

Parity operator

$$\mathbb{P} \, f(x) = f(-x)$$

$$\mathbb{P}^2 f(x) = P\big(P(f(x))\big) = P\big(f(-x)\big) = f(x)$$

$$P^2 = \mathbb{1}$$

we can see that

$$[P, \mathbb{1}] = P \cdot 1 - 1 \cdot P = 0$$

also

$$[P^2, P] = [\mathbb{1}, P] = 0$$

we know that every function is an eigen-state of unit operator '$\mathbb{1}$', but every fun. is not eigen-state for parity operator.

$$\mathbb{P} \, \phi(x) = \begin{cases} \phi(-x) \\ = \alpha \, \phi(x) \end{cases} \qquad P^2 \phi(x) = \alpha^2 \phi(x) = \phi(x)$$

$$\alpha = \pm 1$$

$\alpha = 1$ even function

odd fun, even functions are eigen states of parity operator

$$P f_e(x) = f(-x) = f(x) \qquad \alpha = 1$$

$$P f_o(x) = f(-x) = -f(x) \qquad \alpha = -1 \;\; \text{odd function.}$$

So set of eigen states of P^2 is much bigger than set of eigen functions of parity operator. / eigen state set of $\mathbb{P}$ is subset of spectrum of P^2 or $\mathbb{K}$.

since $[x, p] \neq 0$ they do not share common complete eigen value set. (But they can share few eigen values. even if $[A, B] \neq 0$).

The uncertainty in an observable :-

we know that

$$\langle \hat{A} \rangle = \langle \Psi | \hat{A} | \Psi \rangle \qquad \text{assuming } \langle \Psi | \Psi \rangle = 1$$

$$\langle \hat{A}^2 \rangle = \langle \Psi | A^2 | \Psi \rangle$$

$$(\Delta \hat{A})^2 = \left\langle \left(\hat{A} - \langle A \rangle \right)^2 \right\rangle$$

$$(\Delta \hat{A})^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$$

for position operator $\hat{x}$

$$\langle \hat{x} \rangle = \langle \Psi | \hat{x} | \Psi \rangle$$

$$|\Psi\rangle = \int dx \; \psi(x) |x\rangle$$

$$\langle \Psi | = \int dx' \; \langle x' | \, \psi^*(x')$$

$$\langle \hat{x} \rangle = \int dx \int dx' \; \psi^*(x') \, \psi(x) \; \langle x' | \hat{A} | x \rangle$$

($\rightarrow$ element of A in position basis.)

$$= \int dx \int dx' \; \psi^*(x') \, \psi(x) \; A(x', x)$$

Note :- we know that

$$\psi(x, t) = \int dp \; \langle x | p \rangle \, \tilde{\Psi}(p, t)$$

find $\langle x | p \rangle$?

we know that,

$$\hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar \mathbb{K}$$

let us find matrix element of $[\hat{x}, \hat{p}]$ in position basis.

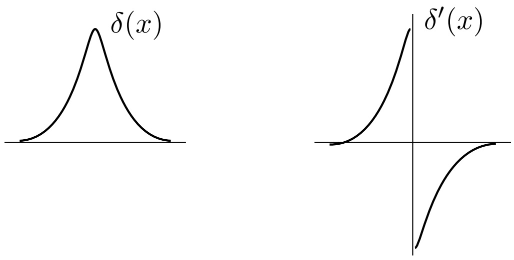
$$\langle x | \hat{x}\hat{p} - \hat{p}\hat{x} | x' \rangle = i\hbar \, \langle x | x' \rangle = i\hbar \, \delta(x - x')$$

$$x^* \langle x | \hat{p} | x' \rangle - x' \langle x | \hat{p} | x' \rangle = i\hbar \, \delta(x - x')$$

(x is real, so $x^* = x$; $\downarrow$ Position has to be real)

$$\langle x | \hat{p} | x' \rangle \, (x - x') = i\hbar \, \delta(x - x')$$

$$\Rightarrow \quad \langle x | \hat{p} | x' \rangle = i\hbar \, \frac{\delta(x - x')}{(x - x')}$$



defined :-

$$\left(\delta'(x) = \frac{-\delta(x)}{x}\right) \qquad \text{(How?)}$$

so

$$\delta'(x-x') = \frac{-\delta(x-x')}{(x-x')}$$

$$\Rightarrow \quad \langle x|\hat{p}|x'\rangle = -i\hbar\,\frac{\partial}{\partial x}\big(\delta(x-x')\big) = -i\hbar\,\frac{\partial}{\partial x}\big(\langle x|x'\rangle\big)$$

for any arbitrary state $\Psi(t)$

$$\langle x|\hat{p}|\Psi(t)\rangle = -i\hbar\,\frac{\partial}{\partial x}\langle x|\Psi(t)\rangle$$

So it $\Rightarrow$

$$\left\{\left(\hat{p} \longrightarrow -i\hbar\frac{\partial}{\partial x}\right) \text{ IN THE POSITION BASIS}\right.$$

$$\hat{x}|x\rangle = x|x\rangle$$

but $\hat{p}|x\rangle$ is totally different $\longrightarrow = -i\hbar\frac{\partial}{\partial x}|x\rangle$

So,

$$\langle x|p|x'\rangle = -i\hbar\,\frac{\partial}{\partial x}\langle x|x'\rangle$$

likewise,

$$\langle x|\hat{x}|x'\rangle = x\,\langle x|x'\rangle = x'\,\langle x|x'\rangle$$

$$\begin{cases} \langle x'|x|x\rangle = 1 & \text{if } x = x' \\ \qquad\qquad = 0 & \text{if } x \neq x' \end{cases} \qquad \langle x|x'\rangle = \delta(x-x')$$

IN Momentum basis :-

$$\hat{p}|p_0\rangle = p_0|p_0\rangle$$

what is $\hat{x}|p_0\rangle =?$

$$\left\{\hat{x} = +i\hbar\frac{\partial}{\partial p} \quad \text{IN momentum Basis}\right.$$

In contineous basis rather than kronical delta we use $\delta(x-x')$.

$$\int f(x')\,\delta(x-x')\,dx' = f(x)$$

$$I = \int \delta(x-x')\,dx' \qquad ?$$

8 Lecture 8

To Find the inner product $\langle x|p\rangle$?

($\langle x| \to$ position eigen state ; $|p\rangle \to$ momentum eigen state)

given $[\hat{x}, \hat{p}] = i\hbar\not{\mathbb{1}}$

what is overlap b/w position eigen states & momentum eigen states. since $[x, p] \neq 0$ we can't find complete common eigen – states. values.

we know that

$$\langle \hat{x}|\hat{p}|\Psi(t)\rangle = -i\hbar\frac{\partial}{\partial x}\langle x|\Psi(t)\rangle$$

($\hat{x} \to$ not a operator)

instead of $\Psi(t)$, we replace it with any momentum eigen -state.

$$\langle \hat{x}|\hat{p}|p\rangle = p\,\langle \hat{x}|p\rangle$$

($\to$ eigen value) as $\hat{p}|p\rangle = \hat{p}|p\rangle \hookrightarrow$ abstract eigen state (regardless of any basis)

also

$$\langle x|\hat{p}|p\rangle = -i\hbar\frac{\partial}{\partial x}\langle x|p\rangle = p\,\langle x|p\rangle$$

$$\langle x|p\rangle = \text{some fun. of } x \text{ labelled by } 'p' = f_p(x)$$

$$-i\hbar\frac{\partial}{\partial x}f_p(x) = p\,f_p(x)$$

$$\int\frac{df(x)}{f(x)} = \int\frac{ip}{\hbar}\,dx$$

$$f_p(x) \propto e^{\frac{ipx}{\hbar}}$$

($p \to$ eigen value corresponding to $\langle x|$)

$$\boxed{\langle x|p\rangle \propto e^{\frac{ipx}{\hbar}}}$$

($\to$ eigen value corresponding to $|p\rangle$)

x, p are both no. (eigen values).

[$\langle x|p\rangle \propto e^{\frac{ip\cdot x}{\hbar}}$, $\langle p|x\rangle \propto e^{-\frac{ip}{\hbar}x}$

here p & x are eigen values of $|p\rangle$ & $|x\rangle$ *respectively.*]

$$\Rightarrow \quad \langle p|x\rangle \sim e^{-\frac{ipx}{\hbar}} \qquad \text{(complex conjugate of } \langle x|p\rangle)$$

we know that

$$\langle x|\Psi(t)\rangle = \Psi(x,t) = \int dp\,\langle x|p\rangle\langle p|\Psi(t)\rangle = \int dp\; e^{\frac{ipx}{\hbar}}\,\tilde{\Psi}(p,t)$$

$$\langle p|\Psi(t)\rangle = \tilde{\Psi}(p,t) = \int dx\,\langle p|x\rangle\langle x|\Psi(t)\rangle = \int dx\; e^{-\frac{ipx}{\hbar}}\,\Psi(x,t)$$

"So we can see that position space wave fun. & momentum space wave functions are fourier transforms of each other."

In 3-D,

$$\langle \vec{r}|\vec{p}\rangle \propto e^{\frac{i\vec{p}\cdot\vec{r}}{\hbar}}$$
$$\langle \vec{p}|\vec{r}\rangle \propto e^{-\frac{i\vec{p}\cdot\vec{r}}{\hbar}}$$

also

$$\langle \vec{r}|\hat{\vec{p}}|\Psi(t)\rangle = -i\hbar\,\nabla_{\vec{r}}\,\langle \vec{r}|\Psi(t)\rangle$$

($\vec{r}, \hat{\vec{p}}$ means 3-D)

The shrödinger equation For a particle :-

$$\hat{H}(\hat{r},\hat{p}) = \hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(\hat{r})$$
$$i\hbar\,\frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

so, we want to know how above eq$^{\text{n}}$ looks like in position basis.

taking inner product with $\langle \vec{r}|\cdots\rangle$ both sides.

$$\left\langle \vec{r}\,\middle|i\hbar\frac{d}{dt}\middle|\,\Psi(t)\right\rangle = \langle \vec{r}|\hat{H}|\Psi(t)\rangle$$

{ Here $|\vec{r}\rangle$ or $\langle\vec{r}|$ is any particular eigen state of position basis.

above eq$^{\text{n}}$ can be written as,

$$i\hbar\,\frac{d}{dt}\big(\langle \vec{r}|\Psi(t)\rangle\big) = \langle \vec{r}|\hat{H}|\Psi(t)\rangle$$
$$i\hbar\,\frac{\partial}{\partial t}\big(\Psi(r,t)\big) = \left\langle \vec{r}\,\middle|\frac{\hat{p}^2}{2m}\middle|\,\Psi(t)\right\rangle + \langle \vec{r}\,|V(\hat{r})|\,\Psi(t)\rangle$$

$\Big\{\frac{d}{dt}$ has to be replaced with $\frac{\partial}{\partial t}$ as derivative is only w.r.t 't' variable.

$\langle \vec{r}|\Psi(t)\rangle$ = position space wave function.

$$\Rightarrow \quad i\hbar\,\frac{\partial}{\partial t}\Psi(r,t) = \frac{(-i\hbar)^2\nabla^2}{2m}\Psi(r,t) + V(\vec{r})\,\psi(r,t)$$
$$\begin{cases} \Psi(r,t) = \langle \vec{r}|\Psi(t)\rangle \\ \quad V(r) = \text{eigen value of } V(\hat{r}) \end{cases}$$

since we deal with hermitian operators in Q.M. (to get real eigen value)

$V(\hat{r})$ should be hermitian too.

So, $V^*(\vec{r})$ is simply $V(\vec{r})$

[complex $V(r)$ is used in case where there is absorption where probabilities are not conserved.]

Find eq$^{\text{n}}$ becomes,

$$\left\{ i\hbar\,\frac{\partial}{\partial t}\Psi(\vec{r},t) = \frac{-\hbar^2}{2m}\nabla^2\Psi(\vec{r},t) + V(\vec{r})\,\Psi(\vec{r},t) \right\}$$

Position space shrödinger equation

Boundary cond^{n}

1) Normalized $\Psi(r,t)$

$$\langle\Psi(t)|\Psi(t)\rangle = 1 = \int d^3r\ |\Psi(\vec{r},t)|^2$$

($\Psi(r,t)$ is member of l_2 space)

Stationary states (eigen states) :-

stationary state of the Q.Mechanical system is by def^{n} is an eigen state of total Hamiltonian of the system.

"Any stationary state is eigen state of $\hat{H}$.

$\hat{H}$ operator has in general infinite no of eigen states".

$$i\hbar\frac{d}{dt}|\Psi(t)\rangle = H|\Psi(t)\rangle$$

lets use $\Phi(t)$ for eigen state (of Hamiltonian) instead of $\Psi(t)$

$$i\hbar\frac{d}{dt}|\Phi(t)\rangle = H|\Phi(t)\rangle = H_0|\Phi(t)\rangle$$

(H_0 is any eigen value of H.)

we know that Hamiltonian tells total energy so lets use E as eigen value of Hamiltonian

E is real eigen value. (as H is made up of physical measurables)

$$i\hbar\frac{d}{dt}|\Phi(t)\rangle = E|\Phi(t)\rangle$$

E = eigen value ; $\Phi(t)$ = eigen state. (any) of hamiltonian

{ there can be more eigen state for one eigen value. for that eigen value should be repeated.

solution to above eq^{n} is

$$\boxed{|\Phi(t)\rangle = e^{-\frac{iEt}{\hbar}}\,|\Phi(0)\rangle}$$

For a general state $|\Psi(t)\rangle$

we can write $|\Psi(t)\rangle$ as linear combination of basis set formed from eigen state of Hamiltonian operator.

$$|\Psi(t)\rangle = \sum_n c_n|\Phi_n(t)\rangle$$

or we can write

$$|\Psi(t)\rangle = \sum_n c_n\, e^{-\frac{iE_nt}{\hbar}}\,|\phi_n(0)\rangle$$

or

$$|\Psi(t)\rangle = \sum_n c_n(t)\,|\phi_n(0)\rangle$$

$$c_n(t) = c_n\, e^{-\frac{iE_n t}{\hbar}}$$

{ Generally we use,

$$|\Psi(t)\rangle = \sum_n c_n\, e^{-\frac{iE_n t}{\hbar}}\, |\phi_n(0)\rangle$$

} The abstract state of any system.

where, $\big(|\phi_n(0)\rangle$ is a choosen time independent basis set$\big)$

$$c_n = \langle\phi_n|\Psi(t)\rangle$$

where, $\{n =$ quantum number$\}$

In General there may be as many quantum no. as there are d.o.f of the system.

for any particle confined in 3-D potential there will be at least 3 quantum no. There may be more if there are internal quantum no. like spin, hypercharge, strangeness for elementry particles.

$$|\Psi(t)\rangle = \sum_n c_n\, e^{-\frac{iE_n t}{\hbar}} |\Phi_n(0)\rangle$$

is also written as

$$\int_n c_n\, e^{-\frac{iE_n t}{\hbar}} |\phi_n(0)\rangle$$

which changes to integral sign when n is contineous.

All quantum interference phenomena happen because of the superposition of different energy states. (Beats, Interference groups, mforms. due to different frequency superposition).

$$|\Psi(t)\rangle = \sum_n c_n e^{-\frac{iE_n t}{\hbar}} |\Phi_n(0)\rangle$$

$\hookrightarrow$ not a fourier series. $\{c_n|\phi_n(0)\rangle$ is aperiodic.

Here we have assumed that all eigen states of Hamiltonian are made orthogonal by Gramh-Schmidt orthonormalization.

lets look for stationary states in position basis $\{|\vec{r}\rangle\}$.

$$i\hbar\frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

taking inner product.

$$\left\langle \vec{r} \middle| i\hbar\frac{d}{dt} \middle| \Psi(t) \right\rangle = \langle\vec{r}|\hat{H}|\Psi(t)\rangle$$
$$= \langle\vec{r}|E\,\Psi(t)\rangle$$

$\{|\vec{r}\rangle$ is any position eigen state$\}$.

$$i\hbar\frac{\partial}{\partial t}\Psi(\vec{r},t) = E\,\Psi(\vec{r},t)$$

$$\Rightarrow \quad \boxed{\Psi(r,t) = e^{-\frac{iEt}{\hbar}}\, |\psi(r,0)\rangle}$$

For a stationary state $\underline{|\psi(r,t)\rangle}$.

For any general state

$$\Psi(r,t) = \sum_n e^{-\frac{iE_n t}{\hbar}} \left|\psi_n(r,0)\right\rangle$$

lets use our old notation $|\phi_n\rangle$ for stationary states.

$$|\phi_n(\vec{r},t)\rangle = e^{-\frac{iE_n t}{\hbar}} \, |\phi_n(\vec{r},0)\rangle$$

So if work in position basis. The $|\Psi(t)\rangle$ can be written as.

$$\langle \vec{r}|\Psi(t)\rangle = \left\langle \vec{r} \middle| \sum_n c_n e^{-\frac{iE_n t}{\hbar}} \middle| \phi_n(\vec{r},0) \right\rangle$$

$$= \sum_n c_n \, e^{-iE_n t/\hbar} \, \langle \vec{r}|\phi_n(\vec{r},0)\rangle$$

where $\langle \vec{r}|\Psi(t)\rangle$ is general state at time 't' in position basis.

So The shrödinger equation for stationary states $|\phi_n(\vec{r},t)\rangle$ looks like

$$\begin{cases} i\hbar \dfrac{\partial}{\partial t}|\phi_n(\vec{r},t)\rangle = \dfrac{(-i\hbar)^2}{2m}\nabla^2\phi_n(\vec{r},t) + V(\vec{r})\,\phi_n(r,t) \\ \qquad\qquad\qquad\quad = E_n\,\phi_n(\vec{r},t) \end{cases}$$

In fact we can remove 't' dependence by only considering

$$\frac{-\hbar^2}{2m}\nabla^2\phi_n(\vec{r},t) + V(\vec{r})\,\phi_n(\vec{r},t) = E_n\,\phi_n(\vec{r},t)$$

where

$$\phi_n(\vec{r},t) = \phi_n(\vec{r},0)\, e^{-\frac{iE_n t}{\hbar}}$$

using above we get,

$$\boxed{\frac{-\hbar^2}{2m}\nabla^2\phi_n(\vec{r},0) + V(\vec{r})\,\phi_n(\vec{r},0) = E_n\,\phi_n(\vec{r},0)}$$

} Time Independent Shrödinger equation for stationary state.

the same equation can be written as :–

$$\left\{ \left(\frac{-\hbar^2}{2m}\nabla^2 + V(\vec{r})\right)|\phi_n(\vec{r})\rangle = E_n|\phi_n(\vec{r})\rangle \right\}$$

eigen value equation. & Homogeneous too.

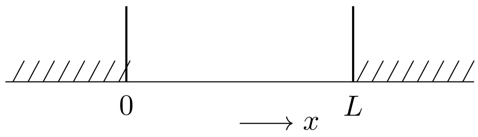
Any confinement on matter "quantizes the energy levels".

A free particle can have any energy but a confined particle in potential / box ensures that only certain packets of energy are allowed.

Q) Where is confinement in H-atom, where e^- is placed in coulomb potential?

Ans.) Hilbert noted that cond$^{\text{n}}$ that $\int |\psi|^2 d^3r = 1$ is enough to produce confinement / quantization

Particle in 1-D BOX :-

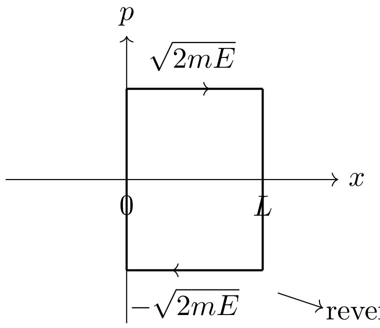


assuming walls are perfect reflectors with no absorption at all.

classically :–

$$H(x,p) = \frac{p^2}{2m} + V(x)$$

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{outside} \end{cases}$$



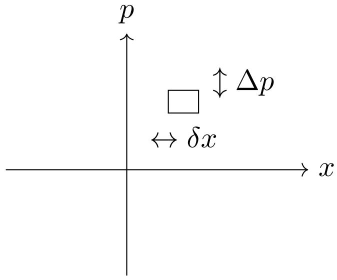
old rule of Bohr's quantization (for periodic orbits) (postulate)

Area under phase space $= \oint p\,dx = nh$

$$\Rightarrow \quad 2\sqrt{2mE_n} \cdot L = nh$$

$$E_n = \frac{n^2 h^2}{8mL^2} \qquad (\text{semi} - \text{classical result}) \qquad \left(\oint P\,dx = nh\right)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$



$\Delta x\, \Delta p \geq \frac{\hbar}{2}$
$(\Delta x\, \Delta p = h)$
(Total area $= nh$)

using time independent shrödinger eq$^{\text{n}}$:–

$$\left[\frac{-\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right] |\phi_n(x)\rangle = E_n |\phi_n(x)\rangle$$

for $(0 < x < L)$ inside the box. $\big(V(x) = 0\big)$

$$\frac{-\hbar^2}{2m}\frac{d^2}{dx^2}|\phi_n(x)\rangle = E_n|\phi_n(x)\rangle$$

$$\left(\frac{d^2}{dx^2} + k_n^2\right)|\phi_n(x)\rangle = 0 \qquad \left\{k_n^2 = \frac{2mE_n}{\hbar^2}\right\}$$

Solutions is trivial

$$|\phi_n(x)\rangle = A\cos k_n x + B\sin k_n x$$

Imposing boundary cond$^{\text{n}}$:–

$$\phi_n(0) = 0 = \phi_n(L)$$

$$\Rightarrow \quad A = 0\ , \qquad k_n L = n\pi$$

as $B \neq 0$

Because of

$$\langle\phi_n(x)|\phi_n(x)\rangle = 1 \quad \text{for all space}$$

$$\int_0^L B^2 \sin^2 k_n x \; dx = 1$$

$$\Rightarrow \quad B^2 \int_0^L \left(\frac{1-\cos 2k_n x}{2}\right) dx = 1$$

$$\Rightarrow \quad B^2\left(\frac{L}{2} - \left(\frac{\sin 2k_n x}{4k_n}\right)_0^L\right) = 1$$

$$B^2\left(\frac{L}{2} - 0\right) = 1 \quad \Rightarrow \quad \boxed{B = \sqrt{\tfrac{2}{L}}}$$

$$|\phi_n(x)\rangle = \sqrt{\frac{2}{L}}\,\sin\left(\frac{n\pi}{L}x\right)$$

and

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2} \qquad (n = 1, 2, 3 \cdots \infty)$$

where Bohr rule also predicts same as exact eigen values.

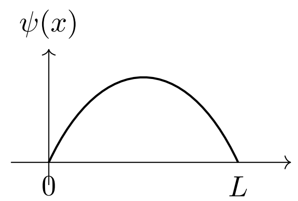
note :– ($n \neq 0$ as $|\phi_n(x)\rangle$ vanishes & normalization does not work)

what is

$$|\phi_n(x,t)\rangle = e^{-\frac{iE_n t}{\hbar}}|\phi_n(x)\rangle \qquad \& \quad e^{-\frac{iE_n t}{\hbar}}\underline{|\phi_n(0)\rangle}$$

Because of lack of symmetry, we can also see that for any Energy eigen value we have a unique eigen state, so there is no degeneracy in this problem.

$$\phi_n(x) = \begin{cases} \sqrt{\frac{2}{L}}\,\sin\left(\frac{n\pi}{L}\right)x & 0 \le x \le L \\ 0 & x < 0,\ x > L \end{cases}$$



[$\psi'(x)$ is disc has finite discontinuity at $x = 0, L$

So, $\psi''(x)$ should have infinite discontinuity at $x = 0, L$, and that discontinuity must be compensated by $V(x)$

$$\left(\frac{-\hbar^2}{2m}\right)\frac{d^2\phi_n}{dx^2} + V(x)\,\phi(x) = E_n\phi(x)$$

]

$$\int_{-\infty}^{\infty} dx\ |\phi_n(x)|^2 = 1$$

$$[\phi_n(x)] = \frac{1}{\sqrt{\text{length}}} \qquad \text{diamensions of } |\phi_n(x)\rangle$$

we do not have following const. in $\phi_n(x)$ due to some reasons.

const.	Reason for absence
c	no relativity
G	no gravity
k_B	assuming $T = 0\,\mathrm{K}$ (absolute)
$\hbar$	

[$\hbar$ **does appear** in wavefunctions of H-atom $R(r) \propto e^{-(r/a_0)}$, $\underline{a_0 = f(\hbar)}$]

sin( ) $\longrightarrow$ must depend on x. & must be dimensionless.

$\left(\sqrt{\frac{2}{L}}\right)$ sin( ) $\longrightarrow$ To normalize.

So '$\hbar$' don't appear in this problem because of dimensional reasons.

lets check if $|\phi_n(x)\rangle$ is momentum eigen state or not.

$$\hat{p}|p\rangle = p|p\rangle \qquad (\rightarrow \text{eigen value})$$

$$-i\hbar\frac{\partial}{\partial x}\phi_n(x) \neq (\ )\,\phi_n(x)$$

$$\Rightarrow \quad |\phi_n(x)\rangle \ \text{ is not a momentum eigen state.}$$

But $[H, \hat{P}] = 0$ as $H = \frac{\hat{p}^2}{2m}$

So we must have some common eigen states.

But we just found out that momentum eigen states are not momentum eigen states.

[$\rightarrow$ Energy / Hamiltonian]

{ So Only solution is "Particle is not free".

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\text{so} \quad \begin{cases} [H, p^2] \neq 0 \\ [H, P] \neq 0 \end{cases}$$

what would $|p\rangle$ (momentum eigen states) looks like?

$$\hat{p}\,\chi(x) = p\,\chi(x)$$

$$\Rightarrow \quad -i\hbar \frac{d\chi}{dx} = p\chi$$

$$\Rightarrow \quad \chi(x) \propto e^{\frac{ipx}{\hbar}}$$

so in position basis, momentum eigen states must look like this.

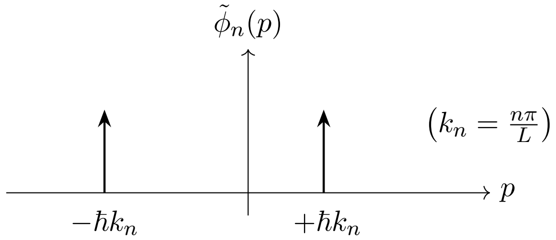
$$\phi_n(x) = \sqrt{\frac{2}{L}} \left(\frac{1}{2i}\right) \left[e^{ik_n x} + e^{-ik_n x}\right]$$

$$\left(k_n = \frac{n\pi}{L}\right)$$

$$\frac{ipx}{\hbar} = \frac{i(n\pi/L)x}{1}$$

$$\boxed{p = \frac{n\pi\hbar}{L}} \qquad \& \quad p = -\frac{n\pi\hbar}{L}$$

Solving problem in momentum space wave function



, delta fun.

[**Q)** we know that if $\phi(x) \in l_2$, $\tilde{\phi}(p)$ must also be a member of l_2. But here dirac functions are not square integrable so whats the solution?

$$\left(\delta(k - k_0) = \int_{-\infty}^{\infty} e^{i(k-k_0)x}\, dx\right)$$

iff $(-\infty < x < \infty)$ there will be $\tilde{\phi}(p) =$ delta funct at $\pm\hbar k_n$

But $0 < x < L$ so $\tilde{\phi}(p)$ is not delta fun of p]

lets find $\tilde{\phi}(p)$

$\tilde{\phi}(p)$ is fourier transform of $\phi(x)$.

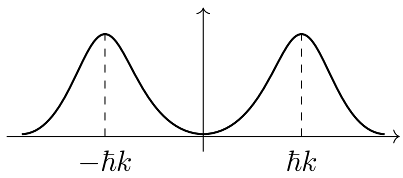
$$\tilde{\phi}(p) = \int_{-\infty}^{\infty} dx\ \phi_n(x)\, e^{-\frac{ipx}{\hbar}}$$

$$= \sqrt{\frac{2}{L}} \int dx\ \frac{e^{-\frac{ipx}{\hbar}}}{2i} \left(e^{ik_n x} - e^{-ik_n x}\right)$$

$$\tilde{\phi}(p) = \sqrt{\frac{2}{L}} \int_0^L \frac{e^{-\frac{ipx}{\hbar}}}{2i} \left(e^{ik_n x} - e^{-ik_n x}\right) dx$$

So,

$$\underline{\tilde{\phi}(p) = \text{smooth function of } p,\ \in l_2}$$



"So particle in box can have any momentum value"

$$\langle A \rangle = \frac{\langle \Psi | A | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

since Ψ is normalized $\Rightarrow \langle \Psi | \Psi \rangle = 1$

$$\langle A \rangle = \langle \Psi | A | \Psi \rangle$$

$\langle A \rangle$ is time independent. Thats why we call it a stationary state.

$$\langle A \rangle = \langle \Psi | A | \Psi \rangle$$

$$= \int dx' \int dx \; \langle \Psi | x \rangle \langle x | A | x' \rangle \langle x' | \Psi \rangle$$

$$\langle \hat{x} \rangle_n = \int dx' \int dx \; \phi_n^*(x) \, \langle x | \hat{x} | x' \rangle \, \phi_n(x)$$

($\rightarrow$ acts upon)

$$\begin{cases} \hat{x}|x'\rangle = x'|x'\rangle \\ \hat{x}|x\rangle = x|x\rangle \end{cases}$$

$$= \int dx' \int dx \; \phi_n^*(x) \; x \; \langle x | x' \rangle \, \phi_n(x)$$

$$= \int dx' \int dx \; \phi_n^*(x) \; x \; \delta(x - x') \, \phi_n(x)$$

doing x' integral ; $\big(\delta(x - x') = 1$ for $x = x'$, $\phi_n(x') = \phi_n(x)$ at $x = x'\big)$

$$\boxed{\langle x \rangle_n = \int dx \; \phi_n^*(x) \; x \; \phi_n(x)}$$

$\hookrightarrow$ calculated in eigen state $\phi_n(x)$.

(where; x = eigen value of $\hat{x}$; $\langle x | \hat{x} = x \langle x |$; $n \rightarrow n^{\text{th}}$ eigen state of Hamiltonian operator)

so,

$$\langle x \rangle_n = \int_0^L dx \; \frac{2}{L} \sin^2\left(\frac{n\pi}{L} x\right) \cdot x$$

$$\langle x^2 \rangle_n = \int_0^L dx \; \frac{2}{L} \sin^2\left(\frac{n\pi}{L} x\right) \cdot x^2$$

compute $(\Delta x)_n$ uncertainty / std. deviation in position.

$$(\delta x)_n = \sqrt{\langle x^2 \rangle_n - \langle x \rangle_n^2}$$

similarly,

$$\langle p\rangle_n = \int_{-\infty}^{\infty} dx\ \phi_n^*(x)\,(-i\hbar)\frac{d}{dx}\phi_n(x)$$

$$\langle p\rangle_n = (-i\hbar)\int_0^L dx\ \frac{2}{L}\sin\left(\frac{n\pi}{L}x\right)\cos\left(\frac{n\pi}{L}x\right)\left(\frac{n\pi}{L}\right) = 0$$

$$\langle p^2\rangle_n = \int_0^L dx\ \frac{2}{L}\sin\left(\frac{n\pi}{L}x\right)\left(\frac{n\pi}{L}\right)^2(-i\hbar)^2\sin\left(\frac{n\pi}{L}x\right)(-1)$$

$$\langle p^2\rangle_n = \frac{2}{L}\cdot\frac{n^2\pi^2\hbar^2}{L^2}\int_0^L \sin^2\left(\frac{n\pi}{L}x\right)dx$$

$$= \frac{n^2\pi^2\hbar^2}{L^2}$$

also

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} = \boxed{\frac{\langle p^2\rangle_n}{2m}}$$

(only iff energy is only K.E.)

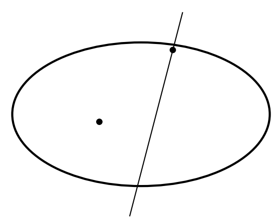
$$(\Delta p)_n = \sqrt{\langle p^2\rangle_n - \langle p\rangle^2} = \frac{n\pi\hbar}{L}$$

$$(\Delta x)_n = \sqrt{\langle x^2\rangle_n - \langle x\rangle_n^2}$$

$$\boxed{(\Delta x)_n\,(\Delta p)_n \geq \hbar}$$

Energy levels are not degenerate in this case.

Particle in a ring :- (circle).



$$\hat{H} = \frac{\hat{p}^2}{2m}$$

in this problem $V(x) = 0$ everywhere, so

$$[H, P] = 0 \quad \text{as } [\hat{p}, p] = 0$$

[1-D simply connected space has no degeneracy, but this space is not simply connected so it may have degeneracy in 1-D. & In fact It has degeneracy of 2. (Its doubly degenerate).]

So every eigen state of H will be a momentum eigen state.

since, $\hat{p}$ and $\frac{\hat{p}^2}{2m}$ has ∞ no of eigen states so They share complete set as, common eigen state.

Periodic Boundary condition :–

$$\phi_n(0) = \phi_n(L)$$
$$\phi_n(x+L) = \phi_n(x)$$

($\phi_n(x)$ is any eigen state of $\hat{H}$.)

If function value are same at x, $x+L$ so their derivative should also be equal.

$$\phi_n'(x+L) = \phi_n'(x)$$
$$\phi_n''(x+L) = \phi_n''(x) \qquad \cdots \text{all derivative must match.}$$

"Are all $|p\rangle$ also eigenstates of $\hat{H}$? (yes) if so, then $|\Psi\rangle = \sum_n c_n|p\rangle$

$$\langle x|\hat{p}|p\rangle = p\,\langle x|p\rangle$$
$$-i\hbar\frac{\partial}{\partial x}\langle x|p\rangle = p\,\langle x|p\rangle \quad \Rightarrow \quad \langle x|p\rangle \propto e^{\frac{ipx}{\hbar}}$$
$$\frac{-\hbar^2}{2m}\frac{\partial^2\phi_n(x)}{\partial x^2} + 0 = E_n\phi_n(x)$$
$$\frac{d^2\phi_n(x)}{dx^2} + \frac{2mE_n}{\hbar^2}\phi_n(x) = 0$$
$$\left(D^2 + k^2\right)\phi_n(x) = 0$$
$$D = \pm ik$$
$$\phi_n(x) = Ae^{ikx} + Be^{-ikx}$$

1)

$$\phi_n(0) = \phi_n(0+L)$$
$$A + B = Ae^{ikL} + Be^{-ikL}$$

2)

$$\phi_n(x) = \phi_n(x+L)$$
$$Ae^{ikx} + Be^{-ikx} = Ae^{ik(x+L)} + Be^{-ik(x+L)}$$
$$Ae^{ikx}\left(1 - e^{ikL}\right) + Be^{-ikx}\left(1 - e^{-ikL}\right) = 0$$
$$\frac{Ae^{ikx}}{Be^{-ikx}} = \frac{e^{-ikL} - 1}{1 - e^{ikL}} \qquad \cdots ①$$

3)

$$\phi_n'(x) = \phi_n'(x+L)$$

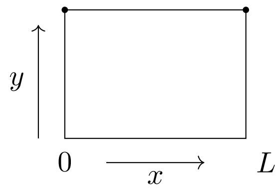
[$\phi_n(x) = Ae^{ikx} + Be^{-ikx}$

$\phi_n(x) = a\cos k_nx + b\sin k_nx$

$$\phi_n(0) = \phi_n(L)$$
$$a = a\cos k_nL + b\sin k_nL$$
$$k_nL = 2n\pi$$
$$k_n = \frac{2n\pi}{L} = \sqrt{\frac{2mE}{\hbar^2}}$$
$$\frac{4n^2\pi^2}{L^2} = \frac{2mE_n}{\hbar^2}$$
$$E_n = \frac{2n^2\pi^2\hbar^2}{mL^2}$$

]

Particle in 2-D box :-



$$\phi_{n_1 n_2}(x,y) = \frac{2}{L}\sin\left(\frac{n_1\pi}{L}\cdot x\right)\sin\left(\frac{n_2\pi}{L}\cdot y\right)$$

$$E(n_1,n_2) = \frac{\hbar^2\pi^2}{2mL^2}\left(n_1^2+n_2^2\right)$$

These energy levels are degenerate except ground state.

$\{$where, $n_1, n_2 = 1, 2, 3\cdots\}$

a) Ground state $\Rightarrow n_1 = 1,\ n_2 = 1$ (non-degenerate).

$$E(1,1) = \frac{\hbar^2\pi^2}{2mL^2}(2)$$

b) First excited state $(1,2)$ or $(2,1)$ degenerate (doubly degenerate)

$$E(1,2) = E(2,1) = \frac{\hbar^2\pi^2}{2mL^2}(5)$$

c) 2^{nd} excited state $(2,2)$ (non-degenerate)

$$E(2,2) = \frac{\hbar^2\pi^2}{2mL^2}(8)$$

d) 3^{rd} excited state $(1,3)$ or $(3,1)$. (doubly degenerate)

if potential were like a ractangle

$$\phi_{n_1 n_2}(x,y) = \frac{2}{\sqrt{L_1 L_2}}\sin\left(\frac{n_1\pi}{L_1}x\right)\sin\left(\frac{n_2\pi}{L_2}x\right)$$

$$E(n_1,n_2) = \frac{\hbar^2\pi^2}{2m}\left(\frac{n_1^2}{L_1^2}+\frac{n_2^2}{L_2^2}\right)$$

Now degeneracy depends on L_1 & L_2 in energy levels.

[The notebook does not carry a separate "Lec 9" heading – the material for it runs on inside the pages transcribed on the Lec 7-8 page. This page begins at the "lec 10" heading in the scans.]

9 Lecture 10

→ "symmetry implies degeneracy".

→ If a problem is integrable in classical mechanics does not imply that same problem will be integrable in quantum mechanically.

ex.

$$H(x,p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + \lambda x^4$$

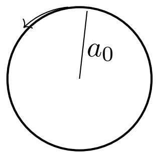
classically Integrable (1 C.O.M requred for 1.D.of $= H$)

But quantum mechanically we can't find exect eigen values of H. we can find it with arbitary accuracy using some perturbation method.

Nature of symmetry in Q.M has very – very deep profound implications as opposed to classical mechanics.

ex In H- atom, we assumed that e^- orbit around nucleous.

$$L \neq 0$$



according to bohr theory

$$\boxed{L \neq 0} \in \; L = mvr = n\hbar \qquad (n = 1, 2, 3)$$

$n =$ angular quantum no.

$$\frac{mv^2}{r} = \frac{Ze^2}{r^2}$$

$$E_n \propto \frac{-1}{n^2} \qquad (n = 1, 2, 3 \cdots)$$

Ground state of H. (1S) (spherically symmetric).

$$n = 1, \; l = 0, \; m = 0$$

eigen fun. can be written as

$$\left\{ \phi_{100}(r, \theta, \phi) \propto e^{-r/a_0} \right.$$

$$l = 0 \quad \Rightarrow \quad L = 0$$

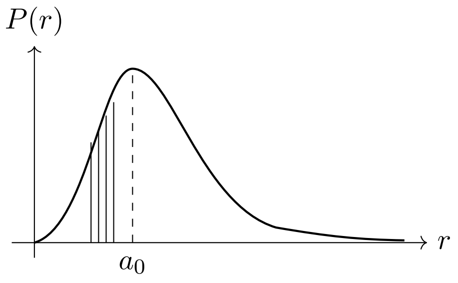
So, How is that quantum theory suggests that angular momentum of e^- in ground state is zero but classical mechanics / intution says its not zero.

The e^- does not orbit with radius a_0. we can find e^- anywhere b/w 0 to ∞ $(0 < r < \infty)$, and probability of finding it b/w r, $r + dr$ is

$$P(r)\,dr = |\phi_{100}(r)|^2 \; dV$$

$$= e^{-\frac{2r}{a_0}} \, (4\pi r^2)\, dr$$

$\{\phi_{100}(r, \theta, \phi) = \phi(r)$ (as no dependence on θ, ϕ)



$\{P(r)$ is maximum at $r = a_0\}$

still It does not answer How $L = 0$.

{classically, This could only happen if e^- passes through center}

How it is that the e^- predominently at $r = a_0$ has zero angular momentum.

Classically different solutions which differ in direction of $\vec{L}$ are related by rotation transformations. we can go from one solution (rotation is in xy plane) to another solution (rotation is in yz plane) by rotation of coordinate axis.

So the group of transformations (rotation group) under which the Hamiltonian is invarient. Each element of that group of rotations takes you from one possible solution to another possible solution.

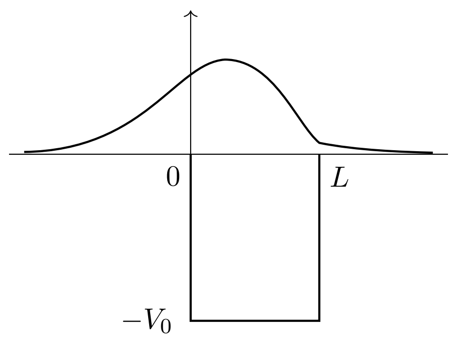
And once you fix coordinate system and specify the initial conditions, the orbit is fixed.

Now Q.M says superposition is valid. So any solution is superposition of all solutions. It says if two solutions are related by a symmetry transformation the general solution is superposition of two solutions.

Now its easy to see that if take the orbit and put it in all possible planes and add all $\vec{L}$. we get $\vec{L} = 0$.

Thats the reason in Q.M we can still sustain a zero $\vec{L}$ solution; even though the e^- has overwhelmingly probability to be at non-zero distance from origin. This is very profound & deep aspect of Q.M.

Particle in Finite well (potential well) :



what are possible values of energy states?

$$\phi_n'' + k_n^2 \phi_n - V(x)\, \phi_n = 0$$

$\hookrightarrow$ finite discountinity. compensated by finite discontinuity of $\phi_n''(x)$.

There is a finite probability that a particle tunnel.

Now,

lets make the problem simpler by assuming that width of the well goes to zero & simultaneously the $V(x) \to \infty$ such that product is finite. we would like to see bound state (i.e. potential can retain the particle).

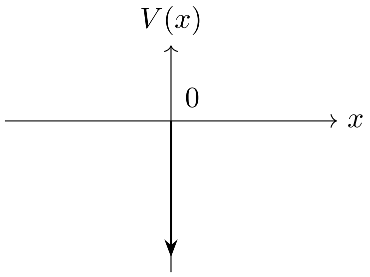
lets assume $V(x)$ is a negative delta function.

Attractive delta – fun. potential :-

$$\frac{-\hbar^2}{2m}\phi''(x) + V(x)\,\phi(x) = E\,\phi(x)$$

$\{\phi(x) =$ eigen state of H. ; $E =$ eigen values of H

lets put δ-fun at $x = 0$, so $V(x) = -\lambda\,\delta(x)$



$$[\lambda] = ML^3T^{-2}$$

because, $[\delta(x)] = L^{-1}$

E must depend on $\lambda, \hbar, m$

$$E \propto \lambda^\alpha\,\hbar^\beta\,m^\gamma$$
$$[ML^2T^{-2}] \propto [ML^3T^{-2}]^\alpha\,[ML^2T^{-1}]^\beta\,m^\gamma$$
$$[M^1L^2T^{-2}] = k\ M^{\alpha+\beta+\gamma}\,L^{3\alpha+2\beta}\,T^{-\beta-2\alpha}$$

$$\begin{array}{ccc} \alpha+\beta+\gamma = 1 & & \beta = -2 \\ 2\beta + 3\alpha = 2 & \Rightarrow & \gamma = +1 \\ -2\alpha - \beta = -2 & & \alpha = 2 \end{array}$$

$$\therefore \quad E \propto \frac{m\lambda^2}{\hbar^2}$$

For $x > 0) \Rightarrow V(x) = 0$

$$\left(\frac{-\hbar^2}{2m}\phi''(x) = E\,\phi(x)\right) \quad \Rightarrow \quad \phi''(x) = -\frac{2mE}{\hbar^2}\phi(x)$$

$\{E < 0$ for bound states so better to write

$$\phi''(x) = +\frac{2m|E|}{\hbar^2}\phi(x)$$

$$\phi''(x) = \frac{2m|E|}{\hbar^2}\phi(x)$$

$$\phi''(x) - k^2\phi(x) = 0 \qquad \left(k = \sqrt{\frac{2m|E|}{\hbar^2}}\right)$$

$$(D^2 - k^2)\phi(x) = 0 \qquad (D = \pm k) \qquad (D = d/dx)$$

$$\phi(x) = Ae^{kx} + Be^{-kx}$$

$$\lim_{x\to\infty} \|\phi(x)\| \longrightarrow \text{finite} \quad \text{so} \quad A = 0$$

$$\phi(x) = Be^{-kx}$$

for $x < 0$)

$$\phi(x) = Ce^{kx} + De^{-kx}$$

normalizability $\Rightarrow D = 0$ [$\lim_{x\to-\infty} \|\phi(x)\| \to$ finite

$$\phi(x) = Ce^{kx}$$

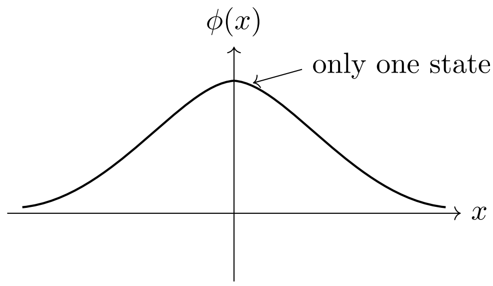
Since $\phi(x)$ is contineous.

$$\phi(x)\Big|_{x\to 0^-} = \phi(x)\Big|_{x=0} = \phi(x)\Big|_{x\to 0^+}$$

$$\Rightarrow \quad \boxed{C = B}$$

So,

$$\phi(x) = Be^{kx}$$



$$\phi'(x(0^-)) \neq \phi'(x(0^+))$$

$$\Rightarrow \quad \phi''(x(0)) \ \text{ has infinite discontinuity.}$$

for $x = 0$)

$$\phi''(x) + \frac{2m\lambda}{\hbar^2}\delta(x)\,\phi(x) = k^2\phi(x) \qquad \left\{k^2 = \frac{-2mE}{\hbar^2}\right\}$$

Integrating from $x = -\epsilon$ to $x = +\epsilon$ & let $\epsilon \to 0$

$$\int_{-\epsilon}^{\epsilon} dx \left\{\phi''(x) + \frac{2m\lambda}{\hbar^2}\delta(x)\,\phi(x)\right\} = \int_{-\epsilon}^{\epsilon} k^2\phi(x)\,dx$$

$$\left(\frac{d\phi(x)}{dx}\right)_{0^-}^{0^+} + \frac{2m\lambda}{\hbar^2}\phi(0) = k^2\left(\int_0^{\epsilon} Be^{-kx}dx + \int_{-\epsilon}^{0} Be^{kx}dx\right) = 0$$

$$\left(\left.\frac{d\phi}{dx}\right|_{x\to 0^+} - \left.\frac{d\phi}{dx}\right|_{x\to 0^-}\right) + \frac{2m\lambda}{\hbar^2}B = 0 \qquad \big(\phi(0) = B\big)$$

$$\begin{cases} x > 0,\ \phi(x) = Be^{-kx}, & \phi'(x) = -kBe^{-kx} \\ x < 0,\ \phi(x) = Be^{kx}, & \phi'(x) = +kBe^{+kx} \end{cases}$$

$$-2kB + \frac{2m\lambda}{\hbar^2}B = 0$$

$$B\left(2k - \frac{2m\lambda}{\hbar^2}\right) = 0$$

$$B \neq 0, \qquad k = \frac{m\lambda}{\hbar^2}\ ; \qquad k^2 = \frac{2m|E|}{\hbar^2} = \frac{m^2\lambda^2}{\hbar^4}$$

$$\Rightarrow \quad |E| = \frac{m\lambda^2}{2\hbar^2}$$

So, for bound state $E < 0$

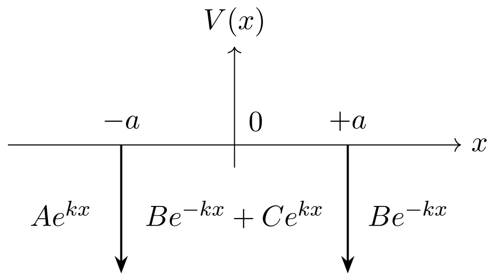
Note :–

$$\boxed{E = \frac{-m\lambda^2}{2\hbar^2}}$$

(only one bound state) as no quantum no exists.

So a single point of ∞-discontinuity in potential supports one B. state

WHAT IF we have two such delta – potential wells



$$V(x) = -\lambda\left[\delta(x-a) + \delta(x+a)\right]$$

{5 eq$^{\text{n}}$ unknown (A, B, C, D, k)}

$\phi(-a)$ is contineous & differentiable $\Rightarrow$ 2 equations

$\phi(+a) \Rightarrow$ 2 equations ; (normalization)

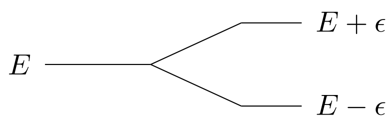
For two states which are independent

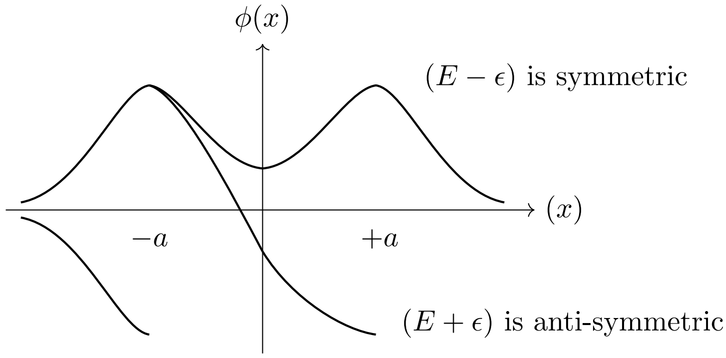
$$H \sim \begin{pmatrix} E & 0 \\ 0 & E \end{pmatrix} \qquad \text{in some vector space.}$$

now, if we switch on the coupling b/w the two

$$H \sim \begin{pmatrix} E & \varepsilon \\ \varepsilon & E \end{pmatrix}$$

eigen values $\Rightarrow E_{1,2} = E \pm \varepsilon$





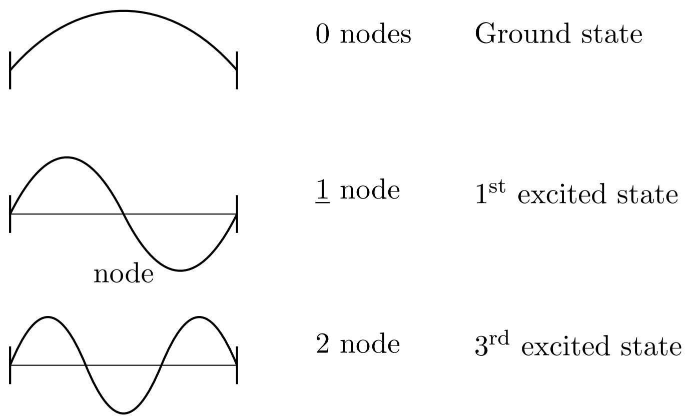
10 Lecture 11

In 1-D problems there is a theorem which says,

Ground state has no node, 1st excited state has one node 2nd excited state has 2 nodes.

so Question is, why does the no. of nodes increase as Energy increases.

To have wave fun. normalizable. It should be clamped at both ends.

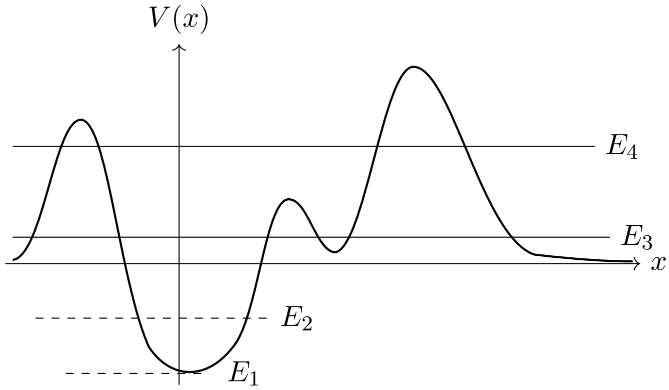


It is because of curvature keeps changing as energy increases. $\phi''(x)$ contributes more and more as curvature changes.

$$\frac{-\hbar^2}{2m}\nabla^2\phi(x) = E\,\phi(x)$$

(curvature (1-D))

So we can see that how ∇^2 contributes to increase of Energy eigen value.



let E_i be total energy of system. $\{E_i\} = \{E_1, E_2, E_3, \cdots\}$

($E_1 < 0$) and $E_1 < V(\text{min})$:–

Such case does not exist as we will see that $\langle \text{K.E} \rangle$ is always a real positive no.

$$\left[\frac{\hat{p}^2}{2m} + V(\hat{x})\right] |\phi(x)\rangle = E|\phi(x)\rangle$$

Since x, p does not commute, $[x, p] \neq 0$ we can not have definite value of K.E and P.E simultaneously in the same state. as.

$$\left[\frac{p^2}{2m}, V(x)\right] \neq 0.$$

(or)

In general the eigen states of $\hat{H}$ are not the eigen states of K.E or the P.E separately.

(or)

In a stationary state, the particle can not have definite value either of its K.E or its P.E. But only sum of the two has definite value of measurement.

So we can't say given the total energy so much is K.E and so much is P.E.

In arbitrary state $|\Psi\rangle$.

$\langle \text{K.E} \rangle$ or $\langle \hat{p}^2 \rangle$ is given by

$$\langle \Psi|\hat{p}^2|\Psi\rangle = \langle \Psi|\hat{p}^\dagger \hat{p}|\Psi\rangle$$

(as $\hat{p}$ is hermitian ($\hat{p} = \hat{p}^\dagger$), ($\hat{p}^2 = \hat{p}^\dagger \hat{p}$))

$$= \left\|\hat{p}|\Psi\rangle\right\|^2 \geq 0$$

(length of vector)

so $\langle \text{K.E} \rangle$ is always a positive real no. i.e. $\langle \text{K.E} \rangle \geq 0$.

so we can't have total energy lesser than P.E because that would imply $\langle \text{K.E} \rangle < 0$.

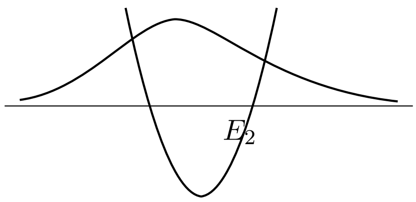
"A particle can not have definite total energy when it is at perticular place. because $[\hat{H}, \hat{x}] \neq 0$.

also,

$$[\hat{H}, \hat{p}] \neq 0 \quad \Rightarrow \quad \text{A particle can't have definite Energy for a given momentum eigen value.}$$

(E_2) $\big(E_2 > V(x)\big)$:

The wave function looks like this :–

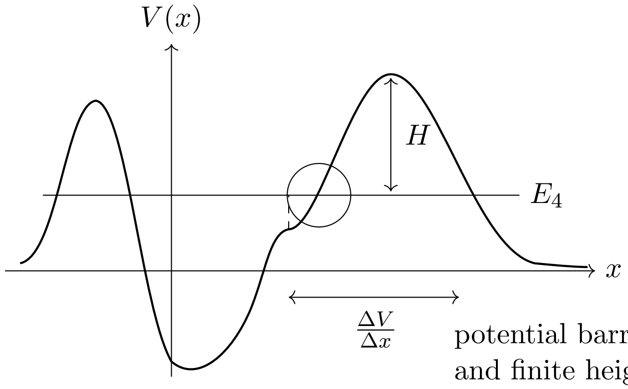


Most of the time we will find particle inside the well, but there is a finite non-zero probability that it can exist outside the well (tunneling).

(E_3) & (E_4) :–

can we have such energy value / level?

Not strictly, These can not be strict stationary states or bound states. Because of the finite width of potential barrier (ΔV), there exists finite probability of the particle to exist outside the barrier / (tunnel through barrier).



so $\exists \int |\Psi(x)|^2 > 0$ for $x >$ barrier width.

so there can't be fixed / bound state.

E_4) If we place particle in such potential. initially. As time goes along this state would evolve under such $\hat{H}$, such that we end up having finite probability to find it outside the well.

Note :–

If we place particle near origin well. and iff we have very large width and height; then we have tunneling probability to be exceedingly low. And we might find the particle inside this well for very – very long time but eventually it is *bound to escape*. Such a state is called metastable state.

Note :- (probability of tunneling)

"Probability of tunneling becomes zero iff the barrier has finite width and infinite height"

$$\lim_{H\to\infty} \big(\text{Tunneling Probability}\big) \longrightarrow 0$$

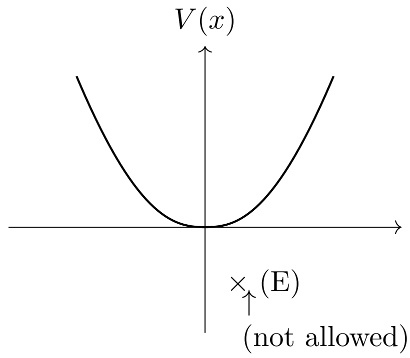
Suppose we have delta fun $(\delta(x))$ barrier, could particle tunnel through it?

Yes because width is infinitensimally small and height is infinite such that their product is finite.

If we have width of barrier infinite from either side, then we can't have E_2 energy level. But still it should have finite probability to tunnel to nearby finite barrier wells. (?)(?)(?).

Question :- What happens if potential has certain symmetry ?

How it is reflected in the eigen states / eigen functions of $\hat{H}$.



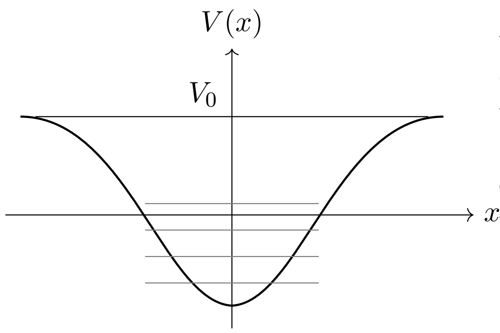
$V(x) \neq V(-x)$ say,

So what kind of eigen states or energy eigen value you would expect?

($E < 0$ not allowed)

$E > 0$ allowed. for (stationary / eigen states of $\hat{H}$) and discrete set of energy eigen values.

For we such potential



we will have (for $E < V_0$) discrete set of eigen values (in the well) and contineous set of eigen values outside the well. ($E > V_0$)

For $E > V_0$ eigen functions are non – normalizable.

Each eigen function must obey :–

$$\frac{-\hbar^2}{2m}\frac{d^2\phi}{dx^2} + V(x)|\phi(x)\rangle = E|\phi(x)\rangle$$

let $x' = -x$, or $(x = -x')$

eq^{n} becomes.

$$\frac{-\hbar^2}{2m}\frac{d^2(\phi(x'))}{dx'^2} + V(-x')\,\phi(-x') = E\,\phi(-x')$$

that would the proved $V(-x') = V(x')$ (or odd)

$$\frac{-\hbar^2}{2m}\frac{d^2}{dx'^2}|\phi(-x')\rangle + V(x')|\phi(-x')\rangle = E|\phi(-x')\rangle$$

$\Rightarrow$ if $\phi(x)$ is solution with corresponding eigen value E so is $\phi(-x)$.

"since there is no degeneracy in 1-D, so every eigen value should have unique eigen fun / state.

$$\Rightarrow \quad \phi(-x) = c\,\phi(x)$$

$\{|\Psi\rangle$ & $\alpha|\Psi\rangle$ are same thing in Q.M

or

$$\phi(x) = c\,\phi(-x)$$
$$\Rightarrow \quad \phi(x) = c\big(c(\phi(x))\big)$$
$$\phi(x)\left[1 - c^2\right] = 0 \quad \Rightarrow \quad \boxed{c = \pm 1}$$

so,

$$\phi(-x) = \pm\,\phi(x)$$
$$\phi(-x) = \begin{cases} \phi(x), & \text{a even function} \\ -\,\phi(x), & \phi(x) \text{ is odd function} \end{cases}$$

If potential has symmetry (reflection / mirror symmetry) / Parity invariance then solutions have definite parity. (either odd / even parity).

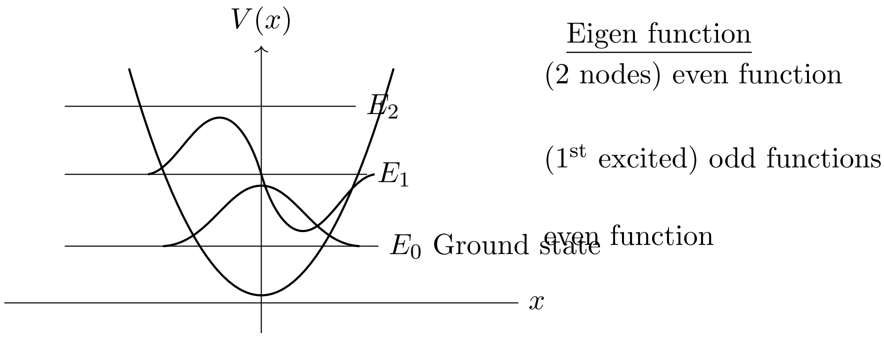
If potential is even function, wave function must be even or odd, but it can't be a mixed function.

$$[\hat{H}, \hat{P}] = 0 \qquad \hat{P} = \text{Parity operator}$$

What would the ground state be? (even or odd)?

Even because, if it were odd, wave fun must be contineous and because it is an odd fun it must have zero value at $x = 0$, so it means it has node. Ground state does not have node.

so ground state will be even function.



$$[\hat{H}, \hat{P}] = 0$$

$\Rightarrow$ All eigen functions of $\hat{H}$ are eigen functions of parity operator.

But converse is not true (All odd/even functions are not eigen fun. of this perticular $\hat{H}$). (simultaneously.

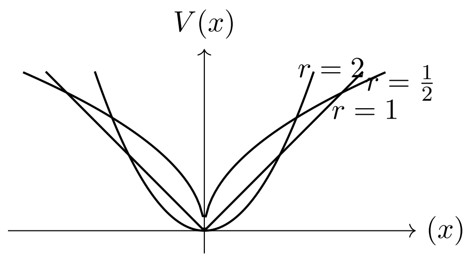
eigen fun. of $\hat{H}$ are subset of eigen fun. of Parity operator.

Question :– what kind of eigen functions / eigen values we expect?

let the general

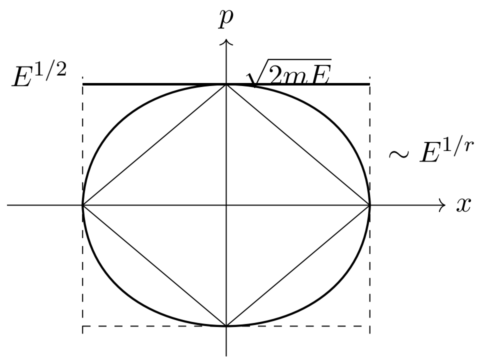
$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = \frac{\hat{p}^2}{2m} + \lambda|x|^r$$

$\hookrightarrow$ for it to be ($V(x)$ even fun.)



In all of such potential we expect bound state with discrete energy levels. and wave functions to be eigen odd / even functions

classical (phase trajectories looks like)



lets find out semiclassically.

lets find,

$$\oint P\,dx = nh$$

so, area goes like

$$E^{\frac{1}{2}+\frac{1}{r}} \in \left(E^{\frac{1}{2}} \cdot E^{\frac{1}{r}}\right)$$

$$\oint P\,dx \sim E^{\frac{r+2}{2r}} \sim nh$$

$$E^{\frac{r+2}{2r}} \sim n$$

$$\boxed{E_n \sim n^{\frac{2r}{r+2}}}$$

for large quantum no (n) we expect results of quantum theory to be matched with semiclassical result.

for $r = 2$

$$\boxed{E_n \sim n\hbar\omega}$$

so we expect that for $n \gg 1$ energy levels are proportional to 'n' itself